

Supplementary Information

PRAgent: Claim-level Scientific Reproduction Across 100 Physics Papers

PRAgent Collaboration

27 August 2026

S1 Claim-first reproduction measurement

The frozen cohort contains 100 papers, 1,427 authored Claims and 3,933 directly mapped Claim Checks. Of the Claims, 1,412 have at least one numerical Check and 15 are handled by a separate formula/invariant or exclusion rule.

For paper p , let C_p be the equal-weight mean, over numerical Claims, of each Claim’s reproduced-Check fraction, expressed in percentage points, and let F_p be trusted scientific-region Fidelity over the measured successful mass on a 0–100 scale. The optional conditional point estimate is $R_p = C_p F_p / 100$. No missing Fidelity is imputed: the tables also report evidence coverage and bounds obtained by assigning every unmeasured successful Check 0 or 100.

Table S1: Corpus-level Claim-first accounting. Fidelity is conditional on trusted evidence and is not a success criterion by itself.

Quantity	Claim Check	Equal-weight Claim
Final resolution (%)	100.0	100.0
Successful reproduction (%)	52.6	40.5
Objective blocker (%)	28.8	21.9
Attempted, not reproduced (%)	18.6	37.5
Pending (%)	0.0	0.0
Conditional Fidelity	92.2	93.0
Fidelity evidence coverage (%)	23.3	23.4
Conditional Degree estimate	48.5	37.7
Degree bounds	11.3–51.6	8.8–39.9

At Check level the mutually exclusive counts are 2,068 reproduced, 1,134 objectively blocked, 731 attempted but not reproduced and 0 pending.

S1.1 Per-paper results

Table S2: Claim-first metrics for all 100 papers. Res., Succ., Block and Fail are percentages. Fid. cov. is the successful-Claim mass with trusted Fidelity. The Degree column is the conditional point estimate; the final column is the no-imputation bound.

Paper ID	N_C	Res.	Succ.	Block	Fail	Fid. cov.	Fid.	Degree	Bound
0709.0548	17	100.0	76.5	23.5	0.0	0.0	–	–	0.0–76.5
0911.0556	4	100.0	75.0	25.0	0.0	60.0	92.3	69.2	41.5–71.5
10.1007-s11433-024-2478-8	14	100.0	54.8	0.0	45.2	0.0	–	–	0.0–54.8
10.1038-s41467-025-67768-4	8	100.0	62.5	37.5	0.0	0.0	–	–	0.0–62.5
10.1038-s41586-024-08262-7	2	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
10.1038-s41586-026-10720-3	13	100.0	7.7	92.3	0.0	0.0	–	–	0.0–7.7
10.1103-PRXQuantum.6.010331	4	100.0	50.0	50.0	0.0	0.0	–	–	0.0–50.0
10.1103-PhysRevB.91.085420	6	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
10.1103-PhysRevLett.124.113601	4	100.0	20.0	0.0	80.0	100.0	95.9	19.2	19.2–19.2
10.1126-science.aaw1611	5	100.0	82.5	17.5	0.0	0.0	–	–	0.0–82.5
10.1145-3297858.3304023	4	100.0	50.0	50.0	0.0	0.0	–	–	0.0–50.0
1004.0190	10	100.0	80.0	20.0	0.0	0.0	–	–	0.0–80.0
1106.2978	7	100.0	88.1	0.0	11.9	28.4	91.8	80.9	23.0–86.1
1206.2505	15	100.0	66.7	0.0	33.3	40.0	83.4	55.6	22.2–62.2
1212.3324	9	100.0	66.7	33.3	0.0	83.3	87.8	58.6	48.8–59.9
1302.5069	13	100.0	100.0	0.0	0.0	14.0	97.5	97.5	13.7–99.7
1508.03344	7	100.0	14.3	85.7	0.0	100.0	95.9	13.7	13.7–13.7
1608.02589	61	100.0	0.0	86.9	13.1	–	–	0.0	0.0–0.0
1706.07435	5	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
1708.05014	3	100.0	67.2	0.0	32.8	0.8	91.4	61.4	0.5–67.1
1709.03478	3	100.0	0.0	100.0	0.0	–	–	0.0	0.0–0.0
1710.10890	11	100.0	27.3	72.7	0.0	0.0	–	–	0.0–27.3
1711.03528	10	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
1711.08863	4	100.0	100.0	0.0	0.0	25.0	98.8	98.8	24.7–99.7
1711.09418	32	100.0	56.2	0.0	43.8	5.6	95.8	53.9	3.0–56.1
1803.01876	8	100.0	62.5	0.0	37.5	80.0	93.5	58.4	46.7–59.2

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Table S2 continued

Paper ID	N_C	Res.	Succ.	Block	Fail	Fid. cov.	Fid.	Degree	Bound
1803.07128	3	100.0	11.1	0.0	88.9	0.0	—	—	0.0–11.1
1804.03151	1	100.0	94.1	5.9	0.0	0.0	—	—	0.0–94.1
1804.04672	10	100.0	50.0	50.0	0.0	0.0	—	—	0.0–50.0
1805.00931	4	100.0	65.0	35.0	0.0	100.0	90.0	58.5	58.5–58.5
1807.01815	4	100.0	16.7	0.0	83.3	0.0	—	—	0.0–16.7
1807.02414	5	100.0	86.7	13.3	0.0	0.0	—	—	0.0–86.7
1807.03311	2	100.0	50.0	50.0	0.0	14.3	91.4	45.7	6.5–49.4
1807.10084	4	100.0	100.0	0.0	0.0	0.0	—	—	0.0–100.0
1807.10676	39	100.0	2.6	41.0	56.4	0.0	—	—	0.0–2.6
1810.00888	35	100.0	100.0	0.0	0.0	20.0	88.8	88.8	17.8–97.8
1811.08017	1	100.0	100.0	0.0	0.0	100.0	94.0	94.0	94.0–94.0
1812.05561	11	100.0	9.1	90.9	0.0	0.0	—	—	0.0–9.1
1903.05124	5	100.0	100.0	0.0	0.0	0.0	—	—	0.0–100.0
1904.10246	3	100.0	100.0	0.0	0.0	100.0	91.7	91.7	91.7–91.7
1905.09460	4	100.0	65.9	34.1	0.0	37.9	95.1	62.7	23.8–64.7
1910.00020	4	100.0	100.0	0.0	0.0	0.0	—	—	0.0–100.0
1910.08980	17	100.0	0.0	1.2	98.8	—	—	0.0	0.0–0.0
2005.09722	11	100.0	90.9	9.1	0.0	0.0	—	—	0.0–90.9
2005.12667	7	100.0	90.1	9.9	0.0	0.0	—	—	0.0–90.1
2101.12558	5	100.0	0.0	100.0	0.0	—	—	0.0	0.0–0.0
2103.03074	21	100.0	46.0	15.9	38.1	0.0	—	—	0.0–46.0
2105.08076	2	100.0	83.3	0.0	16.7	0.0	—	—	0.0–83.3
2211.15015	5	100.0	6.7	86.7	6.7	0.0	—	—	0.0–6.7
2305.15556	3	100.0	100.0	0.0	0.0	0.0	—	—	0.0–100.0
2401.05830	4	100.0	50.0	0.0	50.0	0.0	—	—	0.0–50.0
2401.08523	5	100.0	100.0	0.0	0.0	100.0	96.2	96.2	96.2–96.2
2402.14814	9	100.0	88.9	11.1	0.0	0.0	—	—	0.0–88.9
2404.08378	7	100.0	57.1	42.9	0.0	0.0	—	—	0.0–57.1
2406.07531	3	100.0	0.0	100.0	0.0	—	—	0.0	0.0–0.0
2406.13410	4	100.0	60.3	39.7	0.0	0.0	—	—	0.0–60.3
2407.01296	30	100.0	45.9	54.1	0.0	0.0	—	—	0.0–45.9
2409.18176	5	100.0	38.7	48.0	13.3	0.0	—	—	0.0–38.7
2412.14271	5	100.0	42.2	0.0	57.8	0.0	—	—	0.0–42.2
2502.20558	285	100.0	10.9	0.4	88.8	0.0	—	—	0.0–10.9
2504.08598	5	100.0	60.0	0.0	40.0	0.0	—	—	0.0–60.0
2506.06669	7	100.0	47.6	0.0	52.4	42.5	93.9	44.7	19.0–46.4
2507.09447	8	100.0	37.5	0.0	62.5	100.0	95.1	35.7	35.7–35.7
2510.12880	9	100.0	11.1	0.0	88.9	0.0	—	—	0.0–11.1
2510.26761	40	100.0	40.0	0.0	60.0	0.0	—	—	0.0–40.0

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Table S2 continued

Paper ID	N_C	Res.	Succ.	Block	Fail	Fid. cov.	Fid.	Degree	Bound
2511.11394	2	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
2512.08279	47	100.0	38.3	0.0	61.7	55.6	97.7	37.4	20.8–37.8
2512.23799	4	100.0	25.0	25.0	50.0	0.0	–	–	0.0–25.0
2602.12212	5	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
2604.08669	6	100.0	0.0	83.3	16.7	–	–	0.0	0.0–0.0
2605.02873	5	100.0	100.0	0.0	0.0	100.0	90.9	90.9	90.9–90.9
2605.25398	7	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
2605.25594	4	100.0	0.0	100.0	0.0	–	–	0.0	0.0–0.0
2606.30255	8	100.0	100.0	0.0	0.0	100.0	97.0	97.0	97.0–97.0
2607.00718	5	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
2607.02157	43	100.0	0.0	100.0	0.0	–	–	0.0	0.0–0.0
2607.08212	7	100.0	14.3	85.7	0.0	0.0	–	–	0.0–14.3
2607.08767	9	100.0	22.2	77.8	0.0	0.0	–	–	0.0–22.2
2607.15070	21	100.0	100.0	0.0	0.0	100.0	97.4	97.4	97.4–97.4
2607.15597	42	100.0	40.5	7.1	52.4	9.5	95.9	38.8	3.7–40.3
2607.22976	4	100.0	100.0	0.0	0.0	100.0	92.6	92.6	92.6–92.6
2607.23978	5	100.0	15.0	0.0	85.0	0.0	–	–	0.0–15.0
2607.27060	5	100.0	100.0	0.0	0.0	100.0	100.0	100.0	100.0–100.0
2607.28795	65	100.0	4.6	6.2	89.2	0.0	–	–	0.0–4.6
2608.03987	3	100.0	0.0	100.0	0.0	–	–	0.0	0.0–0.0
2608.05312	6	100.0	2.8	97.2	0.0	0.0	–	–	0.0–2.8
PhysRevA.47.5138	7	100.0	100.0	0.0	0.0	100.0	88.2	88.2	88.2–88.2
PhysRevB.14.2239	9	100.0	33.3	44.4	22.2	100.0	93.0	31.0	31.0–31.0
PhysRevLett.132.1130014	100.0	45.0	55.0	0.0	0.0	0.0	–	–	0.0–45.0
PhysRevLett.133.1918013	100.0	0.0	100.0	0.0	0.0	–	–	0.0	0.0–0.0
cond-mat-0411737	43	100.0	32.7	67.3	0.0	0.0	–	–	0.0–32.7
cond-mat-0503511	49	100.0	99.8	0.0	0.2	40.7	87.0	86.8	35.3–94.5
cond-mat-0509490	5	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0
cond-mat-0610854	3	100.0	0.0	100.0	0.0	–	–	0.0	0.0–0.0
physics-0206018	2	100.0	0.0	0.0	100.0	–	–	0.0	0.0–0.0
physics-9712001	13	100.0	98.1	0.0	1.9	7.2	88.7	87.0	6.3–97.3
quant-ph-0211074	26	100.0	51.1	6.1	42.8	0.0	–	–	0.0–51.1
quant-ph-0403025	3	100.0	66.7	33.3	0.0	100.0	99.7	66.5	66.5–66.5
quant-ph-0507004	20	100.0	60.0	30.0	10.0	0.0	–	–	0.0–60.0
quant-ph-9709029	11	100.0	100.0	0.0	0.0	0.0	–	–	0.0–100.0

S1.2 Claims without numerical Checks

Pure-theory Claims are adjudicated through formula provenance plus an explicit passed derivation or invariant. Experimental and meta-level Claims are excluded from numerical Fidelity instead of being assigned zero or an invented pixel score.

Table S3: All authored Claims without an in-scope numerical Check.

Paper ID	Claim ID	Kind	Adjudication	Reason
10.1103-PRXQuantum.6.010331	CLM004	pure_theory	verified_by_formula_or_invariant	all formula refs resolve with verified gates and a passed independent check
10.1103-PRXQuantum.6.010331	CLM005	pure_theory	partial_formula_verification	non-verified formula gates: EQ011, EQ013
1706.07435	CLM001	pure_theory	verified_by_formula_or_invariant	all formula refs resolve with verified gates and a passed independent check
1903.05124	CLM001	pure_theory	verified_by_formula_or_invariant	all formula refs resolve with verified gates and a passed independent check
1903.05124	CLM002	pure_theory	verified_by_formula_or_invariant	all formula refs resolve with verified gates and a passed independent check
2005.12667	CLM_APPENDICES	pure_theory	partial_formula_verification	no explicit passed formula/invariant check
2402.14814	CLM_IMAGING	pure_theory	partial_formula_verification	no explicit passed formula/invariant check
2504.08598	CLM001	pure_theory	verified_by_formula_or_invariant	all formula refs resolve with verified gates and a passed independent check
2504.08598	CLM006	experimental_or_external_validation	excluded_from_numeric_fidelity	The Claim explicitly records an unperformed real-device validation; no scientific-region pixel or theoretical numerical Fidelity is defined.
2511.11394	CLM002	pure_theory	partial_formula_verification	no explicit passed formula/invariant check
2511.11394	CLM004	pure_theory	partial_formula_verification	no explicit passed formula/invariant check
2511.11394	CLM005	pure_theory	partial_formula_verification	no explicit passed formula/invariant check
2605.02873	CLM006	meta_or_architectural_claim	excluded_from_numeric_fidelity	The architectural comparison is not a standalone numerical observable in the reproduction scope.
2606.30255	CLM-EXPERIMENT	experimental_measurement	excluded_from_numeric_fidelity	The Claim reports measured coincidence data that cannot be independently regenerated from theory-only reproduction code.

Table S3 continued

Paper ID	Claim ID	Kind	Adjudication	Reason
PhysRevLett.132.1301	CLM-METROLOGY	experimental_measurement	excluded_from_numeric_fidelity	The reported frequency is experimental; checking downstream arithmetic does not reproduce the measurement itself and therefore cannot create numerical Fidelity.

S1.3 Every objectively blocked Claim Check

Table S4: Complete objective-blocker ledger. Direct and root causes are projected from linked execution Targets; code-fault status prevents an implementation defect from being relabelled as an external boundary.

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
0709.0548	doherty_extension_claim	CLM_T009	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0709.0548	dqc1_classical_simulation_evidence	CLM_T015	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
0709.0548	main_fig1_finite_ensemble	CLM_T005	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0709.0548	main_fig1_pseudorandom_convergence	CLM_T012	T012	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0709.0548	mixed_state_bounded_entanglement_ppt_speedup	CLM_T015	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
0709.0548	mixed_state_exponential_speedup_evidence	CLM_T015	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
0709.0548	pseudorandom_generator	CLM_T012	T012	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0709.0548	pure_state_entanglement_speedup_theorem	CLM_T015	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
0709.0548	qubit_discord_true_measure_claim	CLM_T015	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_b_activity_alpha_1_2pi	CLM_MICROMASER	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_b_theta_alpha_1_2pi	CLM_MICROMASER	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_c_activity_alpha_2pi	CLM_MICROMASER	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_c_theta_alpha_2pi	CLM_MICROMASER	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_d_active_distribution	CLM_MICROMASER	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_d_coexistence_distribution	CLM_MICROMASER	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
0911.0556	fig3_d_inactive_distribution	CLM_MICROMASER	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41467-025-67768-4	claim_feedback_source_alignment	CLM_FEEDBACK	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
10.1038-s41467-025-67768-4	main_fig2c_simulation_curves	CLM_FEEDBACK	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
10.1038-s41467-025-67768-4	supp_fig10b_logical_cnot_expectation	CLM_LATTICE_SURGERY	T009	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41467-025-67768-4	supp_fig10c_lattice_surgery_bias_overhead	CLM_LATTICE_SURGERY	T009	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41467-025-67768-4	supp_fig2_qldpc_logical_error_distribution	CLM_QLDPC	T008	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M2a	CLM_T_F2A	T_F2A	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M2b	CLM_T_F2B	T_F2B	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M2c	CLM_T_F2C	T_F2C	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M4a-T	CLM_T_F4A	T_F4A	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M4b-T	CLM_T_F4B	T_F4B	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M4c-T	CLM_T_F4C	T_F4C	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M4d-T	CLM_T_F4D	T_F4D	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M4e-T	CLM_T_F4E	T_F4E	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	M4f-T	CLM_T_F4F	T_F4F	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	S1a	CLM_T_S1A	T_S1A	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	S1b	CLM_T_S1B	T_S1B	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1038-s41586-026-10720-3	S1c	CLM_T_S1C	T_S1C	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig10_theory_decay	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig11_many_body_response	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig12_cavity_transfer	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig12_filtered_projection	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig16b_circuit_sensitivity	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig17_phase_flip_model	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig1c_error_budget	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig1f_error_scaling	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig3_full_model_series	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig4_noise_response	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig5_ssb_sensitivity	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig6c_error_histogram	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig7_error_source_power	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig8_principal_quantum_number	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig9a_frequency_response	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
10.1103-PRXQuantum.6.010331	fig9b_intensity_response	CLM006	T004	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	fig9c_protocol_infidelity	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	table4_model_fidelities	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1103-PRXQuantum.6.010331	table5_shot_noise_simulations	CLM003	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S11	CLM005	T005_S11	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(H)	CLM003	T003_S20H	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(I)	CLM003	T003_S20I	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(J)	CLM003	T003_S20J	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(K)	CLM003	T003_S20K	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(L)	CLM003	T003_S20L	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(M)	CLM003	T003_S20M	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(N)	CLM003	T003_S20N	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(O)	CLM003	T003_S20O	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(P)	CLM003	T003_S20P	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(Q)	CLM003	T003_S20Q	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(R)	CLM003	T003_S20R	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1126-science.aaw1611	Supplement Figure S20(S)	CLM003	T003_S20S	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1145-3297858.3304023	fig7_optimization_objectives	CLM_T003	T003	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1145-3297858.3304023	fig8_gate_depth_tradeoff	CLM_T003	T003	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
10.1145-3297858.3304023	table2_benchmark_results	CLM_T004	T004	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1004.0190	R046	CLM_T010	T010	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1004.0190	R053	CLM_T009	T009	scientific_definition_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
1004.0190	R054	CLM_T009	T009	scientific_definition_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
1212.3324	Main Figure 3(d)	CLM_T005	T005	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1212.3324	Main Figure 6(b)	CLM_T007	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1212.3324	Main Figure 6(c)	CLM_T008	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1508.03344	Main Figure 1 level statistics	CLM_T001	T001	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1508.03344	Main Figure 1 spin-glass inset	CLM_T002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1508.03344	Main Figure 2(b) pi-phase level statistics	CLM_T004	T004	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1508.03344	Main Figure 2(c) operator spectral function	CLM_T005	T005	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1508.03344	Main Figure 2(d) correlator micromotion	CLM_T006	T006	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1508.03344	Quantitative localization and phase claims	CLM_T007	T007	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	C-MAIN- PERSISTENCE-005	CLM_C_MAIN_ PERSISTENCE.005	T014	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1A-DTC-SR	CLM001	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1A-MBL-LR	CLM_F1A_MBL_LR	T006	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1A-MBL-SR	CLM_F1A_MBL_SR	T005	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1B-FWHM-J015	CLM_F1B_FWHM_J015	T008	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1B-PEAK-J000	CLM001	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1B-PEAK-J015	CLM001	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1C-FFT-J000- EPSFAMILY	CLM001	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F1D-FFT-J015- EPSFAMILY	CLM001	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2A-BROAD-L08	CLM_F2A_BROAD_L08	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2A-BROAD-L10	CLM_F2A_BROAD_L10	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2A-BROAD-L12	CLM_F2A_BROAD_L12	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2A-NARROW-L08	CLM002	T002	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2A-NARROW-L10	CLM002	T002	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2A-NARROW-L12	CLM_F2A_NARROW_ L12	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2B-VAR-J003-L10	CLM002	T002	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2B-VAR-J003-L12	CLM_F2B_VAR_J003_L12	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2B-VAR-J005-L10	CLM002	T002	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2B-VAR-J005-L12	CLM_F2B_VAR_J005_L12	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2B-VAR-J010-L10	CLM002	T002	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F2B-VAR-J010-L12	CLM_F2B_VAR_J010_L12	T009	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F4-INSET-DTC-LR	CLM004	T004	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F4-INSET-MBL-LR	CLM_F4_INSET_MBL_ LR	T007	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F4-VAR-J003	CLM004	T004	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F4-VAR-J005	CLM004	T004	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1608.02589	F4-VAR-J007	CLM004	T004	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1709.03478	Supp. Fig. S1(d)	CLM003	T005	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1709.03478	Supp. Fig. S1(e)	CLM003	T005	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1709.03478	Supp. Fig. S1(f)	CLM003	T005	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1709.03478	Supp. Fig. S2 theory	CLM003	T006	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PCLM-SI- CURVATURE-X	CLM_HORIZONTAL_ CURVATURE	T011	scientific_definition_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PCLM-SI- CURVATURE-Y	CLM_HORIZONTAL_ CURVATURE	T011	scientific_definition_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PCLM-SI-IMAGING- SIGMA-THEORY	CLM_IMAGING_ WIDTH_CALIBRATION	T012	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-001-A-A11	CLM_INTERACTION	T001	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-001-A-A12	CLM_INTERACTION	T001	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-001-A-A22	CLM_INTERACTION	T001	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-001-A-DELTA-A	CLM_INTERACTION	T001	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-001-B-NC	CLM_CRITICAL	T002	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-002-C-RATIO- BAND	CLM_POPULATION	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-002-C-RATIO- EQ	CLM_POPULATION	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-003-A-NC- METASTABLE	CLM_CRITICAL	T004	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-003-A-NC- STABLE	CLM_CRITICAL	T004	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-003-B-SIGMA- METASTABLE	CLM_SIZE	T005	scientific_definition_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-003-B-SIGMA- STABLE	CLM_SIZE	T005	scientific_definition_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-003-C-RATIO- BAND	CLM_POPULATION	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-003-C-RATIO- EQ	CLM_POPULATION	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-004-GPE-B56P45	CLM_DYNAMICS	T008	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-004-GPE-B56P64	CLM_DYNAMICS	T008	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-S02-GPE- CONFINED12	CLM_EXPANSION	T007	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1710.10890	PFIG-S02-GPE-FREE	CLM_EXPANSION	T007	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1804.03151	main_fig1c_dft_ displacement_map	CLM_HUBBARD	D001	indispensable_input_ unavailable	publication_ underspecified	not_applicable	at_least_one_required_target_externally_ blocked — none
1804.04672	PCLM-TRIANGLE- GEOMETRY- INDEPENDENCE	CLM010	T012	parameter_values_ unavailable	publication_ underspecified	not_applicable	at_least_one_required_target_externally_ blocked — none
1804.04672	PFIG-001-BLOCH- MINUS	CLM002	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1804.04672	PFIG-001-BLOCH- PLUS	CLM002	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1804.04672	PFIG-001-NONBLOCH	CLM002	T003	parameter_values_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1804.04672	PFIG-001-OBC-NUMERICAL	CLM002	T003	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S02-A-GAP-DATA	CLM004	T005	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S02-A-GAP-FIT	CLM004	T005	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S02-B-GAP-DATA	CLM004	T005	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S02-B-GAP-FIT	CLM004	T005	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S02-C-GAP-DATA	CLM004	T005	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S02-C-GAP-FIT	CLM004	T005	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S03-BLOCH-MINUS	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PFIG-S03-BLOCH-PLUS	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PFIG-S03-DISK-NUMERICAL	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S03-NONBLOCH	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PFIG-S05-A-DENSITY	CLM007	T008	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S05-B-DENSITY	CLM007	T008	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S05-C-DENSITY	CLM007	T008	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S05-D-DENSITY	CLM007	T008	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S06-A-DENSITY	CLM007	T009	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S06-B-DENSITY	CLM007	T009	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S06-C-DENSITY	CLM007	T009	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PFIG-S06-D-DENSITY	CLM007	T009	parameter_values._unavailable	publication._underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1804.04672	PTBL-S01-GAMMA-005	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-010	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-015	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-020	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-025	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-030	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-035	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-040	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-045	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1804.04672	PTBL-S01-GAMMA-050	CLM005	T006	parameter_values._unavailable	publication._underspecified	not_found_after_checks	external_root_cause_publication._underspecified — none
1805.00931	F3-LEFT-T10	CLM_GAP	T003-T10	paper_scale_execution._not_run	compute_capacity._shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1805.00931	F3-LEFT-T11	CLM_GAP	T003-T11	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1805.00931	F3-LEFT-T12	CLM_GAP	T003-T12	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1805.00931	F3-LEFT-T13	CLM_GAP	T003-T13	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1805.00931	F3-LEFT-T14	CLM_GAP	T003-T14	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1805.00931	F3-LEFT-T15	CLM_GAP	T003-T15	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1805.00931	Q-IID- DISTRIBUTION- INDEPENDENCE	CLM_IID_ UNIVERSALITY	T006	scientific_definition_ unavailable	publication_ underspecified	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.02414	Main Fig. 1, diffusive GHD t=10	CLM_DIFFUSIVE.WALL	T001- DIFFUSIVE	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.02414	Main Fig. 1, diffusive GHD t=20	CLM_DIFFUSIVE.WALL	T001- DIFFUSIVE	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.02414	Main Fig. 1, diffusive GHD t=40	CLM_DIFFUSIVE.WALL	T001- DIFFUSIVE	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.02414	Main Fig. 1, tDMRG t=10	CLM_DIFFUSIVE.WALL	T003	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.02414	Main Fig. 1, tDMRG t=20	CLM_DIFFUSIVE.WALL	T003	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.02414	Main Fig. 1, tDMRG t=40	CLM_DIFFUSIVE.WALL	T003	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.03311	supp_fig5a_dft_d00_ bands	CLM_DFT_BAND_ CHARACTERS	D001	indispensable_input_ unavailable	publication_ underspecified	not_applicable	at_least_one_required_target_externally_ blocked — none
1807.03311	supp_fig5b_dft_d01_ bands	CLM_DFT_BAND_ CHARACTERS	D002	indispensable_input_ unavailable	publication_ underspecified	not_applicable	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F14-I06	CLM_A.F14-I06	D003	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F14-I10	CLM_A.F14-I10	D004	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F14-I16	CLM_A.F14-I16	D005	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F14-I23	CLM_A.F14-I23	D006	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F15-KGAP	CLM_A.F15-KGAP	D012	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F15-Z080	CLM_A.F15-Z080	D011	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F15-Z083	CLM_A.F15-Z083	D010	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F15-Z086	CLM_A.F15-Z086	D009	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F15-Z090	CLM_A.F15-Z090	D008	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	A-F15-Z100	CLM_A.F15-Z100	D007	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	M-F03A-14	CLM_M.F03A-14	D001	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	M-F03A-23	CLM_M.F03A-23	D001	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	M-F03A-HIGH56	CLM_M.F03A-HIGH56	D001	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
1807.10676	M-F03A-INSET	CLM_M.F03A-INSET	D001	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	external_root_cause_external_ dependency_unavailable — none
1807.10676	M-F03A-LOW56	CLM_M.F03A-LOW56	D001	external_resource_ unavailable	external_dependency_ unavailable	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1807.10676	M-F03B-KGAP	CLM_M_F03B-KGAP	D002	external_resource._ unavailable	external_dependency._ unavailable	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1a_h0_fidelity	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1a_h0_log._ infidelity_inset	CLM_H0_LOG._ INFIDELITY_INSET	T011	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1a_optimized._ fidelity	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1a_optimized._ log_infidelity_inset	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1b_h0_entropy	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1b_optimized._ entropy	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1b_schmidt._ family	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1c_ansatz._ couplings	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1c_asymptotic._ reference	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig1c_optimized._ couplings	CLM_REVIVAL	T001	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2a_spectral._ overlap_cloud	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_poisson._ reference	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_rflow_k0._ even	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_rflow_kpi._ odd	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_rflow._ poisson_reference	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_rflow._ wigner_dyson_reference	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_spacing._ distribution_k0_even	CLM_SECTOR_LEVEL._ STATISTICS	T012	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_spacing._ distribution_kpi_odd	CLM_SECTOR_LEVEL._ STATISTICS	T012	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig2b_wigner._ dyson_reference	CLM_SCARS	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig3_fsa_raising	CLM_SU2	T003	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig3_hz._ expectation	CLM_SU2	T003	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig3_hz_linear._ reference	CLM_SU2	T003	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig3_hz_spacing._ residual	CLM_SU2	T003	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	main_fig3_su2_reference	CLM_SU2	T003	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	supp_figs1a_ansatz	CLM_SU2	T005	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	supp_figs1a_optimized._ fsa	CLM_SU2	T005	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	supp_figs1a_optimized._ revival	CLM_SU2	T005	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	supp_figs1a_optimized._ rvals	CLM_SU2	T005	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
1812.05561	supp_figs1a_optimized._ trvar	CLM_SU2	T005	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1812.05561	supp_figs3b_short_fit_n32	CLM_SHORT_WINDOW_REVIVAL_FITS	T013	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs3c_power_law_fit	CLM_REVIVAL_SCALE_POWER_LAW	T015	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs3c_turning_points	CLM_REVIVAL	T006	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs4_entropy_ansatz	CLM_SCARS	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs4_entropy_h2	CLM_SCARS	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs4_entropy_pxp	CLM_SCARS	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs4_overlap_ansatz	CLM_SCARS	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs4_overlap_h2	CLM_SCARS	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs4_overlap_pxp	CLM_SCARS	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs5_first_scar_entropy	CLM_SCARS	T008	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs5_second_scar_entropy	CLM_SCARS	T008	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs5_second_scar_linear_fit	CLM_SECOND_SCAR_ENTROPY_FIT	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs5_second_scar_log_fit	CLM_SCARS	T008	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs6a_toy_model	CLM_TOY	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs6b_toy_model	CLM_TOY	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1812.05561	supp_figs6c_toy_model	CLM_TOY	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	FIG003_A_BANDWIDTH	CLM003	T002	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	FIG003_A_CRITICAL_GUIDE	CLM003	T002	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
1905.09460	FIG003_B_D010	CLM003	T002	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	FIG003_B_D020	CLM003	T002	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	FIG003_B_D030	CLM003	T002	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	FIG003_B_D035	CLM003	T002	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	SUPP001_C_H1_GUIDE	CLM002	T006	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
1905.09460	SUPP001_D_H1_GUIDE	CLM002	T006	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
1905.09460	SUPP001_D_NL	CLM002	T006	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1905.09460	SUPP001_D_NR	CLM002	T006	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
1910.08980	S-FIG1-ADVANTAGE	CLM_RQAOA_ADVANTAGE	T001 — T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
2005.09722	QCLM-CROSSOVER-LENGTH	CLM_CROSSOVER_LENGTH	T032	parameter_values_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2005.12667	M-F04-b	CLM_SECH	T030	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2005.12667	M-F04-c	CLM_SECH	T030	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2101.12558	supp_figs2_bulk_ni_y2_z2	CLM_ELECTRONIC_ORIGIN	T018	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_bulk_o_px	CLM_ELECTRONIC_ORIGIN	T018	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_bulk_o_py	CLM_ELECTRONIC_ORIGIN	T018	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_bulk_o_pz	CLM_ELECTRONIC_ORIGIN	T018	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsubsurface_ni_3x2_r2	CLM_ELECTRONIC_ORIGIN	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsubsurface_ni_t2g	CLM_ELECTRONIC_ORIGIN	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsubsurface_ni_y2_z2	CLM_ELECTRONIC_ORIGIN	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsubsurface_o_px	CLM_ELECTRONIC_ORIGIN	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsubsurface_o_py	CLM_ELECTRONIC_ORIGIN	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsubsurface_o_pz	CLM_ELECTRONIC_ORIGIN	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsurface_ni_3x2_r2	CLM_ELECTRONIC_ORIGIN	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsurface_ni_t2g	CLM_ELECTRONIC_ORIGIN	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsurface_ni_y2_z2	CLM_ELECTRONIC_ORIGIN	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsurface_o_px	CLM_ELECTRONIC_ORIGIN	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsurface_o_py	CLM_ELECTRONIC_ORIGIN	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_subsurface_o_pz	CLM_ELECTRONIC_ORIGIN	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_surface_ni_3x2_r2	CLM_ELECTRONIC_ORIGIN	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_surface_ni_t2g	CLM_ELECTRONIC_ORIGIN	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_surface_ni_y2_z2	CLM_ELECTRONIC_ORIGIN	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_surface_o_px	CLM_ELECTRONIC_ORIGIN	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_surface_o_py	CLM_ELECTRONIC_ORIGIN	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2101.12558	supp_figs2_surface_o_pz	CLM_ELECTRONIC_ORIGIN	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2103.03074	S01-PROBABILITY	CLM_S001	T001 — T002 — T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	S02-JOINT-SPLIT	CLM_S002	T001 — T002 — T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	S03-XEB-DEFINITION	CLM_S003	T001 — T002 — T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	S04-PORTER-THOMAS	CLM_S004	T001 — T002 — T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	S05-CONDITIONAL-DEFINITION	CLM_T003	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	TB2-OURS-BATCH	CLM_T001	T001	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	TB2-OURS-SPACE	CLM_T006	T006	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2103.03074	TB2-OURS-TIME	CLM_T006	T006	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2211.15015	F6TOP-VC-LINE-REUSE	CLM_FLOW_REGIMES	T004	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	external_root_cause_compute_capacity_shortfall — none
2211.15015	F7A-SCALED-v008	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v010	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v012	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v014	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v016	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v020	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v025	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v030	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v035	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7A-SCALED-v040	CLM_PECLET_COLLAPSE	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7B-LINEAR-GUIDE	CLM_PECLET_COLLAPSE	T024	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2211.15015	F7B-ONSET-COLLAPSE-POINTS	CLM_PECLET_COLLAPSE	T017	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	Q-S-M2-BASIS	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	Q-S-M4-BASIS	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	Q-S-S10-NODE-SECTORS	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	Q-S-S10-SEPARABILITY	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	Q-S-S11-ALPHA	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	Q-S-S12-COMREL-COUPPLING	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02a-M0-0	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02a-M2-0	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02a-M2-1	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02a-M4-0	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02a-M4-1	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02a-M4-2	CLM_SPECTRUM_MODEL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02b-M0-0	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02b-M2-0	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02b-M2-1	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02b-M4-0	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02b-M4-1	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2402.14814	S-F02b-M4-2	CLM_SPECTRUM_MODEL	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2402.14814	S-F02c-THEORY	CLM_SPECTRUM_MODEL	T011	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	M1-C-MODE-1562	CLM_DEVICE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	M1-C-MODE-781	CLM_DEVICE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	M4-B-HOM-FIT	CLM_HOM	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	QCLM-HOM-WIDTH-BANDWIDTH	CLM_HOM	T018	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	QCLM-POLING-PERIOD	CLM_POLING_PERIOD	T019	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	QCLM-PREDICTED-FIBER-VISIBILITY	CLM_HOM	T014	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	QCLM-PREDICTED-LNOI-VISIBILITY	CLM_HOM	T014	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	S5-B-HOM-VISIBILITY-FIBER	CLM_HOM	T014	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	S5-B-HOM-VISIBILITY-LNOI	CLM_HOM	T014	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	S5-C-COLLECTED-SPDC-SPECTRUM	CLM_HOM	T014	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2404.08378	S7-ELECTRODE-LOSS	CLM_DEVICE	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_lower_eom_ccsd	CLM_CONVERGENCE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_lower_gw_dmft_hf	CLM_CONVERGENCE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_lower_gw_dmft_pbe	CLM_CONVERGENCE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_lower_gw_ibdet_hf	CLM_CONVERGENCE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_lower_gw_ibdet_pbe	CLM_CONVERGENCE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_upper_eom_ccsd	CLM_CONVERGENCE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_upper_gw_dmft_hf	CLM_CONVERGENCE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_upper_gw_dmft_pbe	CLM_CONVERGENCE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_upper_gw_ibdet_hf	CLM_CONVERGENCE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2a_upper_gw_ibdet_pbe	CLM_CONVERGENCE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2b_bottom_eom_ccsd	CLM_CONVERGENCE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2b_bottom_gw_ibdet	CLM_CONVERGENCE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2b_middle_eom_ccsd	CLM_CONVERGENCE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2b_middle_hf_ibdet	CLM_CONVERGENCE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2b_top_eom_ccsd	CLM_CONVERGENCE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig2b_top_hf_dmft	CLM_CONVERGENCE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig3a_g0w0_pbe_bands	CLM_MATERIAL_ACCURACY	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2406.07531	main_fig3a_gw_ibdet_spectral_weight	CLM_MATERIAL_ACCURACY	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig3b_g0w0_pbe_bands	CLM_MATERIAL_ACCURACY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig3b_gw_ibdet_spectral_weight	CLM_MATERIAL_ACCURACY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4a_g0w0_pbe_band	CLM_NA_NONLOCAL	T006	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4a_gw_ibdet_spectral_weight	CLM_NA_NONLOCAL	T006	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4a_pbe_band	CLM_NA_NONLOCAL	T006	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4b_local_minus_full_dos	CLM_NA_NONLOCAL	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4c_bottom_imag_correction	CLM_NA_NONLOCAL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4c_bottom_real_correction	CLM_NA_NONLOCAL	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4c_top_imag_correction	CLM_NA_NONLOCAL	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	main_fig4c_top_real_correction	CLM_NA_NONLOCAL	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	supplement_table_s6	CLM_MATERIAL_ACCURACY	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.07531	supplement_table_s7	CLM_MATERIAL_ACCURACY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-003-A-ENERGY	CLM_NONTHERMAL_ION	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-003-B-RADIAL	CLM_NONTHERMAL_ION	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-003-C-AXIAL	CLM_NONTHERMAL_ION	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-005-B-P-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-005-B-S-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-A-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-B-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-C-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-D-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T011	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-E-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T012	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-F-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T013	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-G-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T014	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2406.13410	PFIG-006-I-F-WAVE	CLM_PARTIAL_WAVE_RESONANCES	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2407.01296	M4-B-INSET-PROFILES	CLM_M4_B_INSET_PROFILES	T002-B	parameter_values_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2407.01296	M4-B-KAPPA	CLM_M4_B_KAPPA	T002-B	parameter_values_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2407.01296	M4-D-1TO2	CLM_M4_D_1TO2	T002-D	parameter_values_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2407.01296	M4-D-1TO3	CLM_M4_D_1TO3	T002-D	parameter_values_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2407.01296	M4-D-2TO1	CLM_M4.D_2TO1	T002-D	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-D-3TO1	CLM_M4.D_3TO1	T002-D	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-E-DELTA02	CLM_M4.E.DELTA02	T002-E	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-AMOEBA	CLM_M4.F.AMOEBA	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-DELTA-000	CLM_M4.F.DELTA.000	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-DELTA-005	CLM_M4.F.DELTA.005	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-DELTA-010	CLM_M4.F.DELTA.010	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-DELTA-015	CLM_M4.F.DELTA.015	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-DELTA-020	CLM_M4.F.DELTA.020	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	M4-F-DELTA-025	CLM_M4.F.DELTA.025	T002-F	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2407.01296	QCLM-SMOOTH-LIMIT	CLM_QCLM.SMOOTH.LIMIT	T016	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	QCLM-UNION-CONJECTURE	CLM_QCLM.UNION.CONJECTURE	T015	target_scope_incomplete	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	S7-A-N0400	CLM002	T009	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	S7-A-N0900	CLM002	T009	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	S7-A-N1600	CLM002	T009	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	S7-B-N0421	CLM002	T009	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	S7-B-N0935	CLM002	T009	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2407.01296	S7-B-N1625	CLM002	T009	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M1b-R.S1	CLM_RESONANT._ RESISTIVITY	T001	parameter_values._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M1c-L.S1	CLM_RESONANT._ RESISTIVITY	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M1c-L.S2	CLM_RESONANT._ RESISTIVITY	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M1c-L.S3	CLM_RESONANT._ RESISTIVITY	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M1c-L.S4	CLM_RESONANT._ RESISTIVITY	T002	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M3-I.S1	CLM_TEMPERATURE	T005	parameter_values._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	M3-I.S2	CLM_TEMPERATURE	T011	parameter_values._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	S6.S1	CLM_KUBO.PARITY	T009	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2409.18176	S6.S2	CLM_KUBO.PARITY	T009	paper_scale_execution._ not_run	compute_capacity._ shortfall	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2502.20558	C-F-B-CHANNEL	CLM_C.F.B.CHANNEL	T028	scientific_definition._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none
2512.23799	fig_runtime_per_shot	CLM_T003	T003	indispensable_input._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2604.08669	F4-INTENSITY-ITER10	CLM_FIG4.INTENSITY._ CONVERGENCE	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after_checks	at_least_one_required_target_externally._ blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2604.08669	F4-INTENSITY-ITER3	CLM_FIG4.INTENSITY_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONVERGENCE		unavailable	underspecified	checks	blocked — none
2604.08669	F4-INTENSITY-ITER5	CLM_FIG4.INTENSITY_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONVERGENCE		unavailable	underspecified	checks	blocked — none
2604.08669	F4-INTENSITY-ITER8	CLM_FIG4.INTENSITY_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONVERGENCE		unavailable	underspecified	checks	blocked — none
2604.08669	F4-PHASE-ITER10	CLM_FIG4.PHASE_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONTINUITY		unavailable	underspecified	checks	blocked — none
2604.08669	F4-PHASE-ITER3	CLM_FIG4.PHASE_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONTINUITY		unavailable	underspecified	checks	blocked — none
2604.08669	F4-PHASE-ITER5	CLM_FIG4.PHASE_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONTINUITY		unavailable	underspecified	checks	blocked — none
2604.08669	F4-PHASE-ITER8	CLM_FIG4.PHASE_	T002	parameter_values_	publication_	not_found_after_	at_least_one_required_target_externally_
		CONTINUITY		unavailable	underspecified	checks	blocked — none
2604.08669	F5-A-P2WGS-ITER10	CLM_FIG5.P2WGS_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		RUNTIME		unavailable	unavailable	checks	blocked — none
2604.08669	F5-A-P2WGS-ITER5	CLM_FIG5.P2WGS_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		RUNTIME		unavailable	unavailable	checks	blocked — none
2604.08669	F5-A-P2WGS-ITER8	CLM_FIG5.P2WGS_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		RUNTIME		unavailable	unavailable	checks	blocked — none
2604.08669	F5-A-PATH-PLANNING	CLM_FIG5.PATH_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		RUNTIME		unavailable	unavailable	checks	blocked — none
2604.08669	F5-B-TOTAL-ITER10	CLM_FIG5.PIPELINE_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		BOTTLENECK		unavailable	unavailable	checks	blocked — none
2604.08669	F5-B-TOTAL-ITER5	CLM_FIG5.PIPELINE_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		BOTTLENECK		unavailable	unavailable	checks	blocked — none
2604.08669	F5-B-TOTAL-ITER8	CLM_FIG5.PIPELINE_	T003	external_resource_	external_dependency_	not_found_after_	at_least_one_required_target_externally_
		BOTTLENECK		unavailable	unavailable	checks	blocked — none
2604.08669	Q-F4-INTENSITY-3-TO-5	CLM_FIG4.INTENSITY_	T002	parameter_values_	publication_	not_found_after_	external_root_cause_publication_
		CONVERGENCE		unavailable	underspecified	checks	underspecified — none
2604.08669	Q-F4-INTENSITY-CONVERGED	CLM_FIG4.INTENSITY_	T002	parameter_values_	publication_	not_found_after_	external_root_cause_publication_
		CONVERGENCE		unavailable	underspecified	checks	underspecified — none
2604.08669	Q-F4-INTENSITY-SCALE	CLM_FIG4.INTENSITY_	T002	parameter_values_	publication_	not_found_after_	external_root_cause_publication_
		CONVERGENCE		unavailable	underspecified	checks	underspecified — none
2604.08669	Q-F4-PHASE-CONSTRAINED	CLM_FIG4.PHASE_	T002	parameter_values_	publication_	not_found_after_	external_root_cause_publication_
		CONTINUITY		unavailable	underspecified	checks	underspecified — none
2604.08669	Q-F5-P2WGS-0P5MS	CLM_FIG5.P2WGS_	T003	external_resource_	external_dependency_	not_found_after_	external_root_cause_external_
		RUNTIME		unavailable	unavailable	checks	dependency_unavailable — none
2604.08669	Q-F5-PATH-PLANNING-5MS	CLM_FIG5.PATH_	T003	external_resource_	external_dependency_	not_found_after_	external_root_cause_external_
		RUNTIME		unavailable	unavailable	checks	dependency_unavailable — none
2605.25594	PCLM-APPC-BOUNDARY-INVARIANCE	CLM_WEAK	T024	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PCLM-IVB-W4-APPROX40	CLM_LOCALIZED	T023	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-A-N	CLM_WEAK	T018	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-A-T	CLM_WEAK	T018	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-A-TS	CLM_ANDERSON	T001	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-B-N	CLM_WEAK	T018	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-B-T	CLM_WEAK	T018	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-B-TS-ANDERSON	CLM_ANDERSON	T001	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none
2605.25594	PFIG-001-B-TS-WEAK	CLM_WEAK	T001	paper_scale_execution_	compute_capacity_	not_found_after_	at_least_one_required_target_externally_
				not_run	shortfall	checks	blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2605.25594	PFIG-017-A-PT	CLM_LOCALIZED	T017	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2605.25594	PFIG-017-B-ED	CLM_LOCALIZED	T017	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2605.25594	PFIG-017-B-PT	CLM_LOCALIZED	T017	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2605.25594	PFIG-017-C-W19P6	CLM_LOCALIZED	T017	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2605.25594	PFIG-017-C-W28P2	CLM_LOCALIZED	T017	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_a1_tfim_memory_ capacity	CLM_FIG2_A1_TFIM_ MEMORY_CAPACITY	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_a1_tfim_nmse	CLM_FIG2_A1_TFIM_ NMSE	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_a1_tfim_predictive_ capacity	CLM_FIG2_A1_TFIM_ PREDICTIVE_ CAPACITY	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_a2_cluster_memory_ capacity	CLM_FIG2_A2_ CLUSTER_MEMORY_ CAPACITY	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_a2_cluster_nmse	CLM_FIG2_A2_ CLUSTER_NMSE	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_a2_cluster_ predictive_capacity	CLM_FIG2_A2_ CLUSTER_ PREDICTIVE_ CAPACITY	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_b1_tfim_ information_dissipation	CLM_FIG2_B1_TFIM_ INFORMATION_ DISSIPATION	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_b1_tfim_irreversible_ work	CLM_FIG2_B1_TFIM_ IRREVERSIBLE_WORK	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_b2_cluster_ information_dissipation	CLM_FIG2_B2_ CLUSTER_ INFORMATION_ DISSIPATION	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_b2_cluster_ irreversible_work	CLM_FIG2_B2_ CLUSTER_ IRREVERSIBLE_WORK	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c1_tfim_memory_ classical_mi	CLM_FIG2_C1_TFIM_ MEMORY_CLASSICAL_ MI	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c1_tfim_memory_ coherence	CLM_FIG2_C1_TFIM_ MEMORY_ COHERENCE	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c1_tfim_predictive_ classical_mi	CLM_FIG2_C1_TFIM_ PREDICTIVE_ CLASSICAL_MI	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c1_tfim_predictive_ coherence	CLM_FIG2_C1_TFIM_ PREDICTIVE_ COHERENCE	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c2_cluster_memory_ classical_mi	CLM_FIG2_C2_ CLUSTER_MEMORY_ CLASSICAL_MI	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c2_cluster_memory_ coherence	CLM_FIG2_C2_ CLUSTER_MEMORY_ COHERENCE	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none
2607.02157	fig2_c2_cluster_ predictive_classical_mi	CLM_FIG2_C2_ CLUSTER_ PREDICTIVE_ CLASSICAL_MI	T001	paper_scale_execution_ not_run	compute_capacity_ shortfall	not_found_after_ checks	at_least_one_required_target_externally_ blocked — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.02157	fig2_c2_cluster_predictive_coherence	CLM_FIG2_C2_CLUSTER_PREDICTIVE_COHERENCE	T001	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_a1_mg_segment	CLM_FIGS1_A1_MG_SEGMENT	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_a2_mg_mean	CLM_FIGS1_A2_MG_MEAN	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_a2_mg_variance	CLM_FIGS1_A2_MG_VARIANCE	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_a_accumulation_factor	CLM_FIGS1_A_ACCUMULATION_FACTOR	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_a_mg_power_spectrum	CLM_FIGS1_A_MG_POWER_SPECTRUM	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_b_tfim_spectrum_family	CLM_FIGS1_B_TFIM_SPECTRUM_FAMILY	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS1_c_cluster_spectrum_family	CLM_FIGS1_C_CLUSTER_SPECTRUM_FAMILY	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_a1_tfim_memory_tau0	CLM_FIGS2_A1_TFIM_MEMORY_TAU0	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_a1_tfim_memory_tau1	CLM_FIGS2_A1_TFIM_MEMORY_TAU1	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_a1_tfim_memory_tau2	CLM_FIGS2_A1_TFIM_MEMORY_TAU2	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_a2_cluster_memory_tau0	CLM_FIGS2_A2_CLUSTER_MEMORY_TAU0	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_a2_cluster_memory_tau1	CLM_FIGS2_A2_CLUSTER_MEMORY_TAU1	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_a2_cluster_memory_tau2	CLM_FIGS2_A2_CLUSTER_MEMORY_TAU2	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_b1_tfim_prediction_h1	CLM_FIGS2_B1_TFIM_PREDICTION_H1	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_b1_tfim_prediction_h2	CLM_FIGS2_B1_TFIM_PREDICTION_H2	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_b1_tfim_prediction_h3	CLM_FIGS2_B1_TFIM_PREDICTION_H3	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_b2_cluster_prediction_h1	CLM_FIGS2_B2_CLUSTER_PREDICTION_H1	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_b2_cluster_prediction_h2	CLM_FIGS2_B2_CLUSTER_PREDICTION_H2	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_b2_cluster_prediction_h3	CLM_FIGS2_B2_CLUSTER_PREDICTION_H3	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_c1_tfim_nmse_h1	CLM_FIGS2_C1_TFIM_NMSE_H1	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_c1_tfim_nmse_h2	CLM_FIGS2_C1_TFIM_NMSE_H2	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_c1_tfim_nmse_h3	CLM_FIGS2_C1_TFIM_NMSE_H3	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_c2_cluster_nmse_h1	CLM_FIGS2_C2_CLUSTER_NMSE_H1	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.02157	figS2_c2_cluster_nmse_h2	CLM_FIGS2_C2_CLUSTER_NMSE_H2	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.02157	figS2_c2_cluster_nmse_h3	CLM_FIGS2_C2_CLUSTER_NMSE_H3	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig3a_zap_accounting	CLM_FIG3A	FIG3A_ZAP	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig3b_zx_no_insert	CLM_ZX_EXTRACTOR_PANELS	ZX_EXTRACTOR_INPUT_GAP	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig3c_mobius_native	CLM_FIG3C	FIG3C_NATIVE	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4a_3sat_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4b_qaoa_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4c_pspin_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4d_hypergraph_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4e_qram_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4f_multiplier_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4g_qft_control_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig4h_ghz_control_fidelity	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5a_3sat_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5b_qaoa_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5c_pspin_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5d_hypergraph_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5e_qram_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5f_multiplier_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5g_qft_control_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig5h_ghz_control_moves	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig6a_3sat_duration	CLM_ROUTING_PROXY_SCALING	ROUTING_PROXY_SCALING	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig6b_pspin_duration	CLM_ROUTING_PROXY_SCALING	ROUTING_PROXY_SCALING	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig6c_qram_duration	CLM_ROUTING_PROXY_SCALING	ROUTING_PROXY_SCALING	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig6d_3sat_compile_time	CLM_ROUTING_PROXY_SCALING	ROUTING_PROXY_SCALING	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig6e_pspin_compile_time	CLM_ROUTING_PROXY_SCALING	ROUTING_PROXY_SCALING	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig6f_qram_compile_time	CLM_ROUTING_PROXY_SCALING	ROUTING_PROXY_SCALING	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.08212	fig7a_3sat_vs_zx_sensitivity	CLM_ROUTING_PROXY_SENSITIVITY	ROUTING_PROXY_SENSITIVITY	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig7b_pspin_vs_zx_sensitivity	CLM_ROUTING_PROXY_SENSITIVITY	ROUTING_PROXY_SENSITIVITY	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig7c_qram_vs_zx_sensitivity	CLM_ROUTING_PROXY_SENSITIVITY	ROUTING_PROXY_SENSITIVITY	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig7d_3sat_vs_zap_sensitivity	CLM_ROUTING_PROXY_SENSITIVITY	ROUTING_PROXY_SENSITIVITY	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig7e_pspin_vs_zap_sensitivity	CLM_ROUTING_PROXY_SENSITIVITY	ROUTING_PROXY_SENSITIVITY	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig7f_qram_vs_zap_sensitivity	CLM_ROUTING_PROXY_SENSITIVITY	ROUTING_PROXY_SENSITIVITY	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8a_3sat_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8b_qaoa_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8c_pspin_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8d_hypergraph_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8e_qram_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8f_multiplier_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8g_qft_control_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig8h_ghz_control_stages	CLM_ROUTING_PROXY	ROUTING_PROXY	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08212	fig9_zx_insert_diagnostic	CLM_ZX_EXTRACTOR_PANELS	ZX_EXTRACTOR_INPUT_GAP	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08767	FIG11	CLM_FIG11	T_FIG11	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2607.08767	FIG5_A	CLM_TWIRL_DIRECTION	F5A_PROXY	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.08767	FIG5_B	CLM_FIG5B	T_FIG5B	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2607.08767	FIG6_A	CLM_FIG6A	T_FIG6A	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2607.08767	FIG6_B	CLM_FIG6B	T_FIG6B	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2607.08767	FIG7	CLM_FIG7	T_FIG7	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2607.08767	FIG8	CLM_FIG8	T_FIG8	indispensable_input_unavailable	publication_underspecified	not_applicable	at_least_one_required_target_externally_blocked — none
2607.15597	TS03-GATE-ERROR-BUDGET	CLM_TS03_GATE_ERROR_BUDGET	T016	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.15597	TS04-MULTIION-TOGGLE-SCALING	CLM_TS04_MULTIIION_TOGGLE_SCALING	T017	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.15597	TS11-FOWLER-PROJECTION	CLM_TS11_FOWLER_PROJECTION	T031	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2607.28795	TABLE01_ATOM_TIMING_SUMMARY	CLM_TABLE01_ATOM_TIMING_SUMMARY	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	external_root_cause_publication_underspecified — none

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Table S4 continued

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2607.28795	TABLE01_HAL_SUMMARY	CLM_TABLE01_HAL_SUMMARY	T017	indispensable.input_unavailable	publication_underspecified	not_applicable	external_root_cause_publication_underspecified — none
2607.28795	TABLE09_LOGICAL_ERROR_RATES	CLM_TABLE09_LOGICAL_ERROR_RATES	T008	indispensable.input_unavailable	publication_underspecified	not_applicable	external_root_cause_publication_underspecified — none
2607.28795	TABLE11_CYCLE_TIMES	CLM_TABLE11_CYCLE_TIMES	T016	indispensable.input_unavailable	publication_underspecified	not_applicable	external_root_cause_publication_underspecified — none
2608.03987	Fig. 8	CLM_COST_LAW	T008	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.03987	Fig. 9(a)	CLM_ORDER_TRANSFER	T009	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.03987	Fig. 9(b)	CLM_ORDER_TRANSFER	T009	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.03987	Table 1 core	CLM_COST_LAW	T010	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.03987	Table 1 extension	CLM_COST_LAW	T011	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.03987	Table 5	CLM_COMPLEXITY_AUDIT	T012	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	claim_coherent_detuning_robustness	CLM_DETUNING	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
2608.05312	claim_drain_geometry_reversal	CLM_DRAIN_GEOMETRY	T005 — T006	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
2608.05312	claim_qcle_lindblad_agreement	CLM_QCLE	T011	parameter_values_unavailable	publication_underspecified	not_applicable	external_root_cause_publication_underspecified — none
2608.05312	claim_size_independent_rescue_sum_rule	CLM_SIZE	T001 — T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
2608.05312	claim_temperature_detailed_balance_signature	CLM_TEMPERATURE	T004 — T010	parameter_values_unavailable — target_scope_incomplete	publication_underspecified — scope_definition_gap	not_found_after_checks	external_root_cause_publication_underspecified — none
2608.05312	claim_transient_bright_valve_signature	CLM_VALVE	T002 — T003 — T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
2608.05312	main_fig1c_combined_efficiency	CLM_SIZE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig1c_dephasing_efficiency	CLM_SIZE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig1c_no_dissipation_efficiency	CLM_SIZE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig1c_rescue_efficiency	CLM_SIZE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2a_dephasing_dark_n16	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2a_dephasing_dark_n32	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2a_dephasing_dark_n4	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2a_rescue_dark_n16	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2a_rescue_dark_n32	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2a_rescue_dark_n4	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2b_dephasing_sink_n16	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2b_dephasing_sink_n32	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2b_dephasing_sink_n4	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2b_rescue_sink_n16	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2608.05312	main_fig2b_rescue_sink_n32	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2b_rescue_sink_n4	CLM_VALVE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2c_rescue_bright_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2c_rescue_cavity_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2c_rescue_dark_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2c_rescue_sink_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2d_dephasing_bright_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2d_dephasing_cavity_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2d_dephasing_dark_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig2d_dephasing_sink_population	CLM_VALVE	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3a_n6_efficiency_difference_map	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3a_n6_zero_boundary	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3b_n6_both_channels	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3b_n6_dephasing_only	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3b_n6_rescue_only	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3c_n64_both_channels	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3c_n64_dephasing_only	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	main_fig3c_n64_rescue_only	CLM_TEMPERATURE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1a_combined_peak_marker	CLM_DRAIN_GEOMETRY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1a_dephasing_peak_marker	CLM_DRAIN_GEOMETRY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1a_rescue_peak_marker	CLM_DRAIN_GEOMETRY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1a_site_n_efficiency_map	CLM_DRAIN_GEOMETRY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1b_dephasing_only_cut	CLM_DRAIN_GEOMETRY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1b_no_dissipation_baseline	CLM_DRAIN_GEOMETRY	T012	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs1b_rescue_only_cut	CLM_DRAIN_GEOMETRY	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs2a_logarithmic_extrapolation	CLM_SIZE	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs2a_logarithmic_fit	CLM_SIZE	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs2a_measured_gap	CLM_SIZE	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
2608.05312	supp_figs2b_dephasing_efficiency	CLM_SIZE	T008	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2608.05312	supp_figs2b_gap_deficit	CLM_SIZE	T008	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs2b_inverse_n._ bottleneck	CLM_SIZE	T008	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs2b_power_law._ fit	CLM_SIZE	T008	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3a_without._ rescue_bright	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3a_without._ rescue_cavity	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3a_without._ rescue_dark	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3a_without._ rescue_sink	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3b_with_rescue._ bright	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3b_with_rescue._ cavity	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3b_with_rescue._ dark	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs3b_with_rescue._ sink	CLM_VALVE	T009	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs4_n6_zero._ boundary_overlay	CLM_TEMPERATURE	T004	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs5a._ phenomenological_time._ trace	CLM_QCLE	T011	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs5a_qcle_time._ trace	CLM_QCLE	T011	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs5b._ phenomenological._ coupling_curve	CLM_QCLE	T011	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_figs5b_qcle._ coupling_curve	CLM_QCLE	T011	parameter_values._ unavailable	publication._ underspecified	not_applicable	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_cavity._ drain_baseline	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_cavity._ drain_large_n	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_cavity._ drain_longer_time	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_site_n._ baseline	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_site_n._ larger_system	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_site_n._ longer_time	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s1_site_n._ stronger_coupling	CLM_DRAIN._ GEOMETRY	T006	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s2_detuning._ 0_mev	CLM_DETUNING	T007	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s2_detuning._ 10_mev	CLM_DETUNING	T007	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s2_detuning._ 20_mev	CLM_DETUNING	T007	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
2608.05312	supp_table_s2_detuning._ 5_mev	CLM_DETUNING	T007	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none
PhysRevB.14.2239	Main Fig. 3 rectangularized L2 spectrum	CLM_T003	T003	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	at_least_one_required_target_externally._ blocked — none

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Table S4 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
PhysRevB.14.2239	Main Fig. 4 rectangularized C2 spectrum	CLM_T004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevB.14.2239	Main Fig. 5 field-smeared spectrum	CLM_T005	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevB.14.2239	Supporting checks: band count, symmetries, trace edges, bounds, printed eigenvalues, and magnetic-period ordering	CLM_T007	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
PhysRevLett.132.113001	TH-1S-BIND	CLM_QED.BINDING	T013	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.132.113001	F3-LINESHAPE	CLM_SPECTRUM	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.132.113001	11-FIELD-FREE	CLM_STARK	T009	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.133.191801	1801m amplification_factor_145	CLM_AMPLIFIED_RESPONSE	T001	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	external_root_cause_external_dependency_unavailable — none
PhysRevLett.133.191801	1801m combined_systematic_uncertainty	CLM_NULL_CONSTRAINT	T006	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	external_root_cause_external_dependency_unavailable — none
PhysRevLett.133.191801	1801m constraint_improvement_factors	CLM_NULL_CONSTRAINT	T007	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	external_root_cause_external_dependency_unavailable — none
PhysRevLett.133.191801	1801m template_filter_sensitivity_30ft	CLM_TEMPLATE_FILTER	T004 — T005	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	external_root_cause_external_dependency_unavailable — none
PhysRevLett.133.191801	1801m_fig2a.theoretical_response_component	CLM_AMPLIFIED_RESPONSE	T001 — T002 — T003	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.133.191801	1801m_fig2b.matched_filter.component	CLM_TEMPLATE_FILTER	T004	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.133.191801	1801m_fig2c.filter_sensitivity_component	CLM_TEMPLATE_FILTER	T005	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.133.191801	1801m_fig4a.gaussian_uncertainty_component	CLM_NULL_CONSTRAINT	T006	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
PhysRevLett.133.191801	1801m_fig4b.new_constraint_component	CLM_NULL_CONSTRAINT	T007	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R001	CLM_R001	T014	external_resource_unavailable	external_dependency_unavailable	not_applicable	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R007	CLM_R007	T015	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R016	CLM_T003.RASHBA	T003	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R019	CLM_T003.RASHBA	T003	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R020	CLM_T003.RASHBA	T003	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R021	CLM_T003.RASHBA	T003	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R022	CLM_T004.TOPOLOGY	T004	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R023	CLM_T004.TOPOLOGY	T004	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R024	CLM_T004.TOPOLOGY	T004	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R025	CLM_T004.TOPOLOGY	T004	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R026	CLM_T004.TOPOLOGY	T004	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R027	CLM_T004.TOPOLOGY	T004	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none

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Table S4 continued

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
cond-mat-0411737	R055	CLM_T003_RASHBA	T003	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R056	CLM_T003_RASHBA	T003	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R057	CLM_R057	T003 — T006	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R058	CLM_T005_FLUX	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R059	CLM_R059	T003 — T006	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R059B	CLM_R059B	T003 — T006	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R060	CLM_R060	T003 — T006	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R073	CLM_R073	T004 — T006	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R074	CLM_R074	T019	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R074B	CLM_R074B	T004 — T006	target_scope_incomplete	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R085	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R086	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R087	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R088	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R089	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R090	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R091	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R092	CLM_T010_BARE_GAP	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R093	CLM_R093	T020	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R094	CLM_R094	T021	external_resource_unavailable	external_dependency_unavailable	not_applicable	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R095	CLM_T011_FIELD_RASHBA	T011	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R096	CLM_T011_FIELD_RASHBA	T011	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R097	CLM_R097	T022	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R098	CLM_T011_FIELD_RASHBA	T011	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0411737	R099	CLM_R099	T023	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0610854	crossing_drift_text_claim	CLM_CROSSING_DRIFT	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0610854	main_fig1_distribution_panel	CLM_LEVEL_DISTRIBUTIONS	T001	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0610854	main_fig2_bottom_panel	CLM_FINITE_SIZE_FLOW	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
cond-mat-0610854	main_fig2_top_panel	CLM_FINITE_SIZE_FLOW	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	at_least_one_required_target_externally_blocked — none
quant-ph-0211074	R031	CLM_XXZ_REGIME_DISCRIMINATION	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none

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Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
quant-ph-0211074	R036	CLM_NONCRITICAL_SCALING_FUNCTION	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
quant-ph-0211074	R038	CLM_NONCRITICAL_SCALING_FUNCTION	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
quant-ph-0211074	R039	CLM_NONCRITICAL_SCALING_FUNCTION	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
quant-ph-0211074	R040	CLM_NONCRITICAL_SCALING_FUNCTION	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
quant-ph-0211074	R065	CLM_MAJORISATION	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	external_root_cause_publication_underspecified — none
quant-ph-0403025	Section VII n=11/n=17 numerical-simulation claim	CLM_GF4_LENGTH_TREND	T004	indispensable_input_unavailable	publication_underspecified	not_found_after_checks	at_least_one_required_target_externally_blocked — none
quant-ph-0507004	exceptional_point_separatrix_density	CLM_T019	T019	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	independent_review_external_publication_underspecified — none
quant-ph-0507004	main_fig1	CLM_T007	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	independent_review_external_publication_underspecified — none
quant-ph-0507004	normal_spacing_variation	CLM_T014	T014	parameter_values_unavailable	publication_underspecified	not_applicable	independent_review_external_publication_underspecified — none
quant-ph-0507004	ordering_error_scaling	CLM_T017	T017	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	independent_review_external_publication_underspecified — none
quant-ph-0507004	super_scar_localization	CLM_T011	T011	parameter_values_unavailable	publication_underspecified	not_applicable	independent_review_external_publication_underspecified — none
quant-ph-0507004	super_scar_wkb_indices	CLM_T012	T012	parameter_values_unavailable	publication_underspecified	not_applicable	independent_review_external_publication_underspecified — none

S1.4 Every attempted but not reproduced Claim Check

Table S5: Complete attempted-failure ledger. These Checks were not objectively blocked and therefore remain agent-side negative outcomes.

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
10.1007-s11433-024-2478-8	Appendix Figure a7(a)	CLM_T008.A7.HYBRID	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a7(b)	CLM_T008.A7.HYBRID	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a7(c)	CLM_T008.A7_AMPLITUDE	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a7(d)	CLM_T008.A7_AMPLITUDE	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a8(a)	CLM_T008.A8.HYBRID	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a8(b)	CLM_T008.A8.HYBRID	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a8(c)	CLM_T008.A8_AMPLITUDE	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Appendix Figure a8(d)	CLM_T008.A8_AMPLITUDE	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Main Figure 4(a)	CLM_T004.TWO_PHOTON_HYBRID.CZ	T004	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Main Figure 4(b)	CLM_T004.TWO_PHOTON_HYBRID.CZ	T004	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Main Figure 4(c)	CLM_T004.TWO_PHOTON_HYBRID.CZ	T004	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
10.1007-s11433-024-2478-8	Main Figure 4(d)	CLM.T005.TWO_PHOTON_AMPLITUDE_CZ	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Main Figure 4(e)	CLM.T005.TWO_PHOTON_AMPLITUDE_CZ	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Main Figure 4(f)	CLM.T005.TWO_PHOTON_AMPLITUDE_CZ	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1007-s11433-024-2478-8	Main Figure 6(b)	CLM.T007.TOFFOLI_CCZ	T007	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	fig2_inset_ipr_mobility_	CLM001	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	fig2_main_state_	CLM001	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	fig3a_aa_susceptibility_	CLM002	T002	scientific_result_mismatch	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	fig4a_photon_number_	CLM003	T003	scientific_result_mismatch	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	fig4b_threshold_	CLM003	T003	scientific_result_mismatch	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	fig4c_escape	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	figs1a_density_profile_	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	figs1b_density_profile_	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	figs1c_density_profile_	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	figs1d_density_profile_	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
10.1103-PhysRevLett.124.113601e	figs1e_density_profile_	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1106.2978	INV-012	CLM.METHOD	T014	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	independent_review_paper_error_candidate — none
1106.2978	Isotropic connected-correlation claim	CLM.CORRELATIONS	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	independent_review_paper_error_candidate — none
1206.2505	finite_g1_corrections	CLM.T015	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
1206.2505	main_fig3_bottom	CLM.T007	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1206.2505	main_fig3_upper_left	CLM.T005	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1206.2505	main_fig3_upper_right	CLM.T006	T006	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1206.2505	supp_fig1	CLM.T008	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1608.02589	C-MAIN-HIDDEN-ISING-MAP	CLM.C.MAIN.HIDDEN_ISING_MAP	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1608.02589	C-MAIN-MEAN-CORR-BETA	CLM.C.MAIN.MEAN_CORR.BETA	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1608.02589	C-MAIN-NU-TYP-AVG	CLM.C.MAIN.NU_TYP_AVG	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1608.02589	C-MAIN-TYP-CORR	CLM.C.MAIN.TYP_CORR	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1608.02589	C-MAIN-WIGNER-DECAY	CLM.C.MAIN.WIGNER_DECAY	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1608.02589	C-SI-HIDDEN-SYMM	CLM.C.SI.HIDDEN_SYMM	T016	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1608.02589	C-SI-RANDOM-ROT	CLM.C.SI.RANDOM_ROT	T016	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1608.02589	C-SI-ZN- UNIVERSALITY	CLM_C_SI_ZN_ UNIVERSALITY	T016	implementation_not_ ready	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
1708.05014	M1-C-N040	CLM_BTC	T001	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M1-C-N080	CLM_BTC	T001	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M1-C-N160	CLM_BTC	T001	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M1-C-NINF	CLM_BTC	T001	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M2-L-INSET-ZOOM	CLM_BTC	T003	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ target_after_attempt — none
1708.05014	M2-L-SPECTRUM	CLM_BTC	T002	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M2-R-INSET-ZOOM	CLM_BTC	T005	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ target_after_attempt — none
1708.05014	M2-R-SPECTRUM	CLM_BTC	T004	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J01	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J02	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J03	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J04	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J05	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J06	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J07	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J08	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J09	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-L-J10	CLM_BTC	T006	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M3-R-BANDS	CLM_BTC	T007	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M4-L-N040	CLM_BTC	T008	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M4-L-N080	CLM_BTC	T008	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M4-L-N160	CLM_BTC	T008	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M4-R-LAMBDA1	CLM_BTC	T010	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	M4-R-LAMBDA2	CLM_BTC	T010	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	S2-L-SX	CLM_BTC	T011	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	S2-L-SY	CLM_BTC	T011	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	S2-L-SZ	CLM_BTC	T011	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	S2-R-SX	CLM_BTC	T012	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none
1708.05014	S2-R-SY	CLM_BTC	T012	paper_scale_execution_ not_run	review_or_evidence_ process_gap	not_found_after_ checks	fresh_independent_review_inconclusive_ after_attempt — none

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Table S5 continued

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1708.05014	S6-R-FIELD	CLM_BTC	T020	parameter_values._ unavailable	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1708.05014	S6-TRAJECTORIES	CLM_BTC	T020	parameter_values._ unavailable	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1708.05014	S7-A	CLM_BTC	T021	parameter_values._ unavailable	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1708.05014	S7-B	CLM_BTC	T022	parameter_values._ unavailable	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1708.05014	S7-C	CLM_BTC	T023	parameter_values._ unavailable	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1708.05014	S7-D	CLM_BTC	T024	parameter_values._ unavailable	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C022	CLM_C022	T004	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C023	CLM_C023	T004	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C024	CLM_C024	T004	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C025	CLM_C025	T004	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C026	CLM_C026	T005	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C027	CLM_C027	T005	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C028	CLM_C028	T005	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C029	CLM_C029	T005	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C030	CLM_C030	T007	parameter_values._ unavailable	publication._ underspecified	not_applicable	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C031	CLM_C031	T005	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C032	CLM_C032	T005	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C033	CLM_C033	T006	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C034	CLM_C034	T006	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1711.09418	C035	CLM_C035	T006	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.01876	C1-zero-mode-end- migration	CLM006_ZERO_MODE_ MIGRATION	T006	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.01876	C2-multiband-winding- sum	CLM007_MULTIBAND_ ADDITIVITY	T007	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.01876	C3-ref49-zero-mode- correction	CLM008_REF49_ CORRECTION	T008	evidence_chain._ incomplete	review_or_evidence._ process_gap	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.07128	claim_phase_squeezing_ universal_separability	CLM_SEPARABILITY	T005	scientific_result._ mismatch	paper_claim_discrepancy	not_found_after._ checks	independent_review_paper_error._ candidate — none
1803.07128	main_fig5_blobs_svc	CLM_KERNEL	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.07128	main_fig5_c15_capacity	CLM_KERNEL	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.07128	main_fig5_c1_capacity	CLM_KERNEL	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.07128	main_fig5_c2_capacity	CLM_KERNEL	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.07128	main_fig5_circles_svc	CLM_KERNEL	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none
1803.07128	main_fig5_moons_svc	CLM_KERNEL	T002	parameter_values._ unavailable	publication._ underspecified	not_found_after._ checks	fresh_independent_review_inconclusive._ after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1803.07128	main_fig6_epoch1_perceptron	CLM_SEPARABILITY	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1803.07128	main_fig6_epoch5000_perceptron	CLM_SEPARABILITY	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1803.07128	main_fig6_epoch500_perceptron	CLM_SEPARABILITY	T003	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1803.07128	main_fig8_loss_series	CLM_VARIATIONAL	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1803.07128	main_fig8_probability_map	CLM_VARIATIONAL	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 1(a)	CLM_TDVP	T_FIG1A	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 1(b)	CLM_DYNAMICS	T_FIG1B	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 2(a)	CLM_CHAOS	T_FIG2A	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 2(b)	CLM_DYNAMICS	T_FIG2B	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 2(c)	CLM_DYNAMICS	T_FIG2C	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 4(a)	CLM_TDVP	T_FIG4A	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 4(b)	CLM_DYNAMICS	T_FIG4B	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 4(c)	CLM_TDVP	T_FIG4C	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. 4(d)	CLM_DYNAMICS	T_FIG4D	unresolved	unresolved	not_excluded	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. S2(a)	CLM_DEFORMED	T_FIGS2A	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1807.01815	Fig. S2(b)	CLM_DEFORMED	T_FIGS2B	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1807.10676	A-F05A-K	CLM_A_F05A-K	T016	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-F07B-01	CLM_A_F07B-01	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-F07B-02	CLM_A_F07B-02	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-F07B-03	CLM_A_F07B-03	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-F07B-04	CLM_A_F07B-04	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-F07B-05	CLM_A_F07B-05	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-F07B-06	CLM_A_F07B-06	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	A-T05-WINDING	CLM_A_T05-WINDING	A001	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-C3-W2-ADDITIVE	CLM_Q_C3-W2-ADDITIVE	A002	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-FRAGILE-EBR	CLM_Q_FRAGILE-EBR	A002	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-HIGHER-02-05	CLM_Q_HIGHER_02-05	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-HIGHER-06-07	CLM_Q_HIGHER_06-07	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-HIGHER-08-15	CLM_Q_HIGHER_08-15	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-HIGHER-16-17	CLM_Q_HIGHER_16-17	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1807.10676	Q-HIGHER-18-22	CLM_Q_HIGHER_18_22	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-HIGHER-23-24	CLM_Q_HIGHER_23_24	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-HIGHER-25-27	CLM_Q_HIGHER_25_27	T015	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-NESTED-WILSON	CLM_Q_NESTED_WILSON	A002	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-PH-ALGEBRA	CLM_Q_PH_ALGEBRA	A002	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-PH-BREAK-ROBUSTNESS	CLM_Q_PH_BREAK_ROBUSTNESS	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-PH-IRREPS	CLM_Q_PH_IRREPS	A002	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1807.10676	Q-STABLE-Z2	CLM_Q_STABLE_Z2	A002	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
1910.08980	C-APP-A-L1-DEPTH	C-APP-A-L1-DEPTH	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-A-L2-GAUGE	C-APP-A-L2-GAUGE	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-B-L1-GHZ	C-APP-B-L1-GHZ	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-B-L2-PRINTED-ENERGY	C-APP-B-L2-PRINTED-ENERGY	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-C-L1-CORRELATION	C-APP-C-L1-CORRELATION	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-C-RUNTIME-O-N4	C-APP-C-RUNTIME-O-N4	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-D-L1-PRINTED-ENERGY	C-APP-D-L1-PRINTED-ENERGY	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-D-L2-ENDPOINT-ENERGY	C-APP-D-L2-ENDPOINT-ENERGY	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-APP-D-L2-RQAOA-1D	C-APP-D-L2-RQAOA-1D	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-C1-QAOA-BOUND	C-MAIN-C1-QAOA-BOUND	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-T1-NLTS	C-MAIN-T1-NLTS	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-T2-RANGE-BOUND	C-MAIN-T2-RANGE-BOUND	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-T2-TIGHTNESS	C-MAIN-T2-TIGHTNESS	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-T3-LOCAL	C-MAIN-T3-LOCAL	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-T3-QAOA	C-MAIN-T3-QAOA	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	C-MAIN-T3-RQAOA	C-MAIN-T3-RQAOA	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	F1-L-QAOA	CLM_RQAOA_ADVANTAGE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	F1-L-RQAOA	CLM_RQAOA_ADVANTAGE	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	F1-R-QAOA	CLM_RQAOA_ADVANTAGE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	F1-R-RQAOA	CLM_RQAOA_ADVANTAGE	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
1910.08980	S-APP-A-T1-RESTATE	C-MAIN-T1-NLTS	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
1910.08980	S-APP-B-L2-TIGHTNESS-PROOF	C-MAIN-T2-TIGHTNESS	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
1910.08980	S-APP-D-L1-QAOA	C-MAIN-T3-QAOA	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
1910.08980	S-APP-D-L1-RQAOA	C-MAIN-T3-RQAOA	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2103.03074	C01-HEAD-TAIL-FACTOR	CLM_T005	T005	target_scope_incomplete	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C02-BATCH-REUSE-SCALING	CLM_T005	T005	target_scope_incomplete	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C03-20C-PARTITION	CLM_T005	T005	target_scope_incomplete	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C06-PEAK-MEMORY	CLM_T013	T013	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	independent_review_paper_error_candidate — none
2103.03074	C07-COMPLEX64-PRECISION	CLM_T009	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C13-MARGINAL-XEB-IDENTITY	CLM_T012	T012	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	independent_review_paper_error_candidate — none
2103.03074	C15-NOISY-FIDELITY-COST	CLM_T014	T014	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C16-FULL43-STATE	CLM_T015	T015	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C17-FULL43-PARTITION	CLM_T015	T015	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C18-FULL43-RUNTIME	CLM_T015	T015	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C19-FULL50-STATE	CLM_T016	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C20-FULL50-PARTITION	CLM_T016	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C21-FULL50-BATCH-COST	CLM_T016	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C22-FULL50-PT-HIST	CLM_T016	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C23-FULL50-RUNTIME	CLM_T016	T016	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2103.03074	C24-FOLLOWON-1M-SAMPLES	CLM_T017	T017	external_resource_unavailable	external_dependency_unavailable	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2105.08076	Main Figure 3(b)	CLM_DARK_STATE	T007	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2211.15015	C003-DT-CONVERGENCE	CLM_VALIDATION	T027	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2211.15015	C004-SIZE-CONVERGENCE	CLM_VALIDATION	T028	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2401.05830	claim_dissipator_rate_factor_two	CLM_PRINTED_RATE_DISCREPANCY	T011	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	independent_review_paper_error_candidate — none
2401.05830	claim_hamiltonian_prose_factor_two	CLM_HAMILTONIAN_PROSE_DISCREPANCY	T012	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	independent_review_paper_error_candidate — none
2409.18176	M4a.S1	CLM_AC_THREE_FLUID	T006	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2409.18176	M4a.S2	CLM_AC_THREE_FLUID	T006	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2409.18176	M4a.S3	CLM_AC_THREE_FLUID	T006	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2409.18176	M4a.S4	CLM_AC_THREE_FLUID	T006	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2409.18176	M4c.S1	CLM_AC_THREE_FLUID	T008	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2409.18176	M4c.S2	CLM_AC_THREE_FLUID	T008	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2409.18176	M4c.S3	CLM_AC_THREE_FLUID	T008	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2409.18176	M4c.S4	CLM_AC_THREE_FLUID	T008	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2412.14271	Formal Fig. S2 / arXiv v1 Fig. 6(a)	CLM002	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Formal Fig. S2 / arXiv v1 Fig. 6(b)	CLM002	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Formal Fig. S2 / arXiv v1 Fig. 6(c)	CLM002	T005	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Formal Fig. S3 (panel inventory unavailable)	CLM005	T006	external_resource_unavailable	external_dependency_unavailable	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Formal Fig. S4 (panel inventory unavailable)	CLM005	T006	external_resource_unavailable	external_dependency_unavailable	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Formal Fig. S5 / arXiv v1 Fig. 7	CLM003	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(a)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(b)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(c)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(d)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(e)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(f)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 3(g)	CLM002	T002	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 4(a)	CLM002	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 4(b)	CLM002	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 4(c)	CLM002	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 4(d)	CLM002	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 4(e)	CLM002	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2412.14271	Main Fig. 4(f)	CLM002	T003	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2502.20558	F10-A-P0P01	CLM_F10_A_P0P01	T012	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F10-B-P0P03	CLM_F10_B_P0P03	T012	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B000	CLM_F11A_B000	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B010	CLM_F11A_B010	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B020	CLM_F11A_B020	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B030	CLM_F11A_B030	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B040	CLM_F11A_B040	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B050	CLM_F11A_B050	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B060	CLM_F11A_B060	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F11A-B070	CLM_F11A_B070	T013	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2502.20558	F12-DC1-ETA100	CLM_F12_DC1.ETA100	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-DC1-ETA50	CLM_F12_DC1.ETA50	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-DC1-ETA500	CLM_F12_DC1.ETA500	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-TELE-ETA0P5	CLM_F12_TELE.ETA0P5	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-TELE-ETA10	CLM_F12_TELE.ETA10	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-TELE-ETA100	CLM_F12_TELE.ETA100	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-TELE-ETA50	CLM_F12_TELE.ETA50	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F12-TELE-ETA500	CLM_F12_TELE.ETA500	T014	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D05-P1	CLM_F15A_D05.P1	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D05-P10	CLM_F15A_D05.P10	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D05-P100	CLM_F15A_D05.P100	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D05-P5	CLM_F15A_D05.P5	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D07-P1	CLM_F15A_D07.P1	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D07-P10	CLM_F15A_D07.P10	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D07-P100	CLM_F15A_D07.P100	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15A-D07-P5	CLM_F15A_D07.P5	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L050-PERR0P01	CLM_F15B_L050_PERR0P01	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L050-PERR0P1	CLM_F15B_L050_PERR0P1	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L050-PERR0P5	CLM_F15B_L050_PERR0P5	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L050-PERR1	CLM_F15B_L050_PERR1	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L050-PERR1P5	CLM_F15B_L050_PERR1P5	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L050-PERR2	CLM_F15B_L050_PERR2	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L100-PERR0P01	CLM_F15B_L100_PERR0P01	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L100-PERR0P1	CLM_F15B_L100_PERR0P1	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L100-PERR0P5	CLM_F15B_L100_PERR0P5	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L100-PERR1	CLM_F15B_L100_PERR1	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L100-PERR1P5	CLM_F15B_L100_PERR1P5	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F15B-L100-PERR2	CLM_F15B_L100_PERR2	T015	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F16A-SWAP1-DATA	CLM_F16A_SWAP1_DATA	T016	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F16A-SWAP1-MEASURE	CLM_F16A_SWAP1_MEASURE	T016	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

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Table S5 continued

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2502.20558	F23-DC2-D07	CLM_F23_DC2_D07	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-DC2-D09	CLM_F23_DC2_D09	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-DC2-D11	CLM_F23_DC2_D11	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-DC2-D13	CLM_F23_DC2_D13	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-DC2-THRESHOLD	CLM_F23_DC2_THRESHOLD	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-SWAP1-D07	CLM_F23_SWAP1_D07	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-SWAP1-D09	CLM_F23_SWAP1_D09	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-SWAP1-D11	CLM_F23_SWAP1_D11	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-SWAP1-D13	CLM_F23_SWAP1_D13	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-SWAP1-THRESHOLD	CLM_F23_SWAP1_THRESHOLD	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-TELE-D07	CLM_F23_TELE_D07	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-TELE-D09	CLM_F23_TELE_D09	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-TELE-D11	CLM_F23_TELE_D11	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-TELE-D13	CLM_F23_TELE_D13	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F23-TELE-THRESHOLD	CLM_F23_TELE_THRESHOLD	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC025-BETA-FIT	CLM_F24_DC025_BETA_FIT	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC025-D07-DATA	CLM_F24_DC025_D07_DATA	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC1-BETA-FIT	CLM_F24_DC1_BETA_FIT	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC1-D07-DATA	CLM_F24_DC1_D07_DATA	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC1-LMI-BETA-FIT	CLM_F24_DC1_LMI_BETA_FIT	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC1-LMI-D07-DATA	CLM_F24_DC1_LMI_D07_DATA	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC2-BETA-FIT	CLM_F24_DC2_BETA_FIT	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-DC2-D07-DATA	CLM_F24_DC2_D07_DATA	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-SWAP1-BETA-FIT	CLM_F24_SWAP1_BETA_FIT	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-SWAP1-D07-DATA	CLM_F24_SWAP1_D07_DATA	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-TELE-BETA-FIT	CLM_F24_TELE_BETA_FIT	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F24-TELE-D07-DATA	CLM_F24_TELE_D07_DATA	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC025-D07	CLM_F25_DC025_D07	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC025-D09	CLM_F25_DC025_D09	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC025-D11	CLM_F25_DC025_D11	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2502.20558	F25-DC025-D13	CLM_F25_DC025_D13	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC025-THRESHOLD	CLM_F25_DC025_THRESHOLD	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-D07	CLM_F25_DC1_D07	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-D09	CLM_F25_DC1_D09	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-D11	CLM_F25_DC1_D11	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-D13	CLM_F25_DC1_D13	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-LMI-D07	CLM_F25_DC1_LMI_D07	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-LMI-D09	CLM_F25_DC1_LMI_D09	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-LMI-D11	CLM_F25_DC1_LMI_D11	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-LMI-D13	CLM_F25_DC1_LMI_D13	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-LMI-THRESHOLD	CLM_F25_DC1_LMI_THRESHOLD	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC1-THRESHOLD	CLM_F25_DC1_THRESHOLD	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC2-D07	CLM_F25_DC2_D07	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC2-D09	CLM_F25_DC2_D09	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC2-D11	CLM_F25_DC2_D11	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC2-D13	CLM_F25_DC2_D13	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-DC2-THRESHOLD	CLM_F25_DC2_THRESHOLD	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-SWAP1-D07	CLM_F25_SWAP1_D07	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-SWAP1-D09	CLM_F25_SWAP1_D09	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-SWAP1-D11	CLM_F25_SWAP1_D11	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-SWAP1-D13	CLM_F25_SWAP1_D13	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-SWAP1-THRESHOLD	CLM_F25_SWAP1_THRESHOLD	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-TELE-D07	CLM_F25_TELE_D07	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-TELE-D09	CLM_F25_TELE_D09	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-TELE-D11	CLM_F25_TELE_D11	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-TELE-D13	CLM_F25_TELE_D13	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F25-TELE-THRESHOLD	CLM_F25_TELE_THRESHOLD	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F26-DC025-BETA-FIT	CLM_F26_DC025_BETA_FIT	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F26-DC025-D07-DATA	CLM_F26_DC025_D07_DATA	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F26-DC1-BETA-FIT	CLM_F26_DC1_BETA_FIT	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2502.20558	F26-DC1-D07-DATA	CLM_F26_DC1_D07_ DATA	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-DC1-LMI-BETA-FIT	CLM_F26_DC1_LMI_ BETA_FIT	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-DC1-LMI-D07-DATA	CLM_F26_DC1_LMI_D07_ DATA	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-DC2-BETA-FIT	CLM_F26_DC2_BETA_ FIT	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-DC2-D07-DATA	CLM_F26_DC2_D07_ DATA	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-SWAP1-BETA-FIT	CLM_F26_SWAP1_ BETA_FIT	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-SWAP1-D07-DATA	CLM_F26_SWAP1_D07_ DATA	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-TELE-BETA-FIT	CLM_F26_TELE_BETA_ FIT	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F26-TELE-D07-DATA	CLM_F26_TELE_D07_ DATA	T020	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F2B-LOSS-EXACTTIME	CLM_F2B_LOSS_ EXACTTIME	T001 — T007	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F2B-LOSS-NOSSR	CLM_F2B_LOSS_NOSSR	T001 — T007	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F2B-LOSS-SSR	CLM_F2B_LOSS_SSR	T001 — T007	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3D-CONV	CLM_F3D_CONV	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3D-DC1	CLM_F3D_DC1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3D-DC1-LMI	CLM_F3D_DC1_LMI	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3D-SWAP1	CLM_F3D_SWAP1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3D-TELE	CLM_F3D_TELE	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3E-CONV	CLM_F3E_CONV	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3E-DC1	CLM_F3E_DC1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3E-DC1-LMI	CLM_F3E_DC1_LMI	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3E-SWAP1	CLM_F3E_SWAP1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3E-TELE	CLM_F3E_TELE	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3F-DC1	CLM_F3F_DC1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3F-DC1-LMI	CLM_F3F_DC1_LMI	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3F-SWAP1	CLM_F3F_SWAP1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3F-TELE	CLM_F3F_TELE	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3G-DC1	CLM_F3G_DC1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3G-DC1-LMI	CLM_F3G_DC1_LMI	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3G-SWAP1	CLM_F3G_SWAP1	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none
2502.20558	F3G-TELE	CLM_F3G_TELE	T008	scientific_result_ mismatch	system_capability_limit	not_found_after_ checks	at_least_one_required_target_attempt_ failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2502.20558	F4A-DC1	CLM_F4A_DC1	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4A-DC1-LMI	CLM_F4A_DC1_LMI	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4A-DC2	CLM_F4A_DC2	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4A-DC3	CLM_F4A_DC3	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4A-DC4	CLM_F4A_DC4	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4A-SWAP1	CLM_F4A_SWAP1	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4A-TELE	CLM_F4A_TELE	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-DC1	CLM_F4B_DC1	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-DC1-LMI	CLM_F4B_DC1_LMI	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-DC2	CLM_F4B_DC2	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-DC3	CLM_F4B_DC3	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-DC4	CLM_F4B_DC4	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-SWAP1	CLM_F4B_SWAP1	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F4B-TELE	CLM_F4B_TELE	T009	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5B-L000	CLM_F5B_L000	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5B-L020	CLM_F5B_L020	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5B-L040	CLM_F5B_L040	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5B-L060	CLM_F5B_L060	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5B-L080	CLM_F5B_L080	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5C-L000	CLM_F5C_L000	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5C-L020	CLM_F5C_L020	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5C-L040	CLM_F5C_L040	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5C-L060	CLM_F5C_L060	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F5C-L080	CLM_F5C_L080	T010	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F7B-ERASURE	CLM_F7B_ERASURE	T011	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F7B-LOSS-EXACTTIME	CLM_F7B_LOSS_EXACTTIME	T011	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F7B-LOSS-NOSSR	CLM_F7B_LOSS_NOSSR	T011	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	F7B-LOSS-SSR	CLM_F7B_LOSS_SSR	T011	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2502.20558	TBL1-DEFF-C1	CLM_TBL1_DEFF_C1	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2502.20558	TBL1-DEFF-C2	CLM_TBL1_DEFF_C2	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2502.20558	TBL1-THRESH-C1	CLM_TBL1_THRESH_C1	T019	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2502.20558	TBL1-THRESH-C2	CLM_TBL1_THRESH_C2	T020	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2504.08598	FIG007	CLM007	T004	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2504.08598	FIG008	CLM004	T003A	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2504.08598	FIG009	CLM004	T003B	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	claim_fst_isospectral_bell_gauge	CLM_FST_BELL_LOCAL_GAUGE	C004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	claim_large_m_effective_half_chain	CLM_LARGE_M_EFFECTIVE_HALF_CHAIN	C003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	claim_zigzag_pst_all_m	CLM_ZIGZAG_PST_ALL_INTEGER_M	C002	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	main_fig4d_theory_q11	CLM_REMOTE.W	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	main_fig4d_theory_q13	CLM_REMOTE.W	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	main_fig4d_theory_q31	CLM_REMOTE.W	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	main_fig4d_theory_q33	CLM_REMOTE.W	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	main_fig4f_ideal_w_density	CLM_REMOTE.W	T007	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2506.06669	supp_figs10a_fidelity_curve	CLM_REMOTE.W	D001	scientific_definition_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs1a_offdiagonal_theory_density	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs1b_offdiagonal_obc_density	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs1c_offdiagonal_profiles	CLM004	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs2a_quasiperiodic_theory_density	CLM005	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs2b_quasiperiodic_obc_density	CLM005	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs2c_quasiperiodic_profiles	CLM005	T005	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs3_precision_energy_3_20	CLM007	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs3_precision_energy_minus_0.72	CLM007	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs3_precision_vs_size_dependence	CLM007	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs3_qr_more_accurate_at_64bit	CLM007	T007	paper_scale_execution_not_run	compute_capacity_shortfall	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs4_exponential_decay_claim	CLM008	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	figs4_gap_deviation_series	CLM008	T008	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2507.09447	main_unidirectional_density_limit	CLM006	T006				fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Claim: open-chain exact-point degeneracy	CLM005	V003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Claim: open-chain $\theta=3\pi/2$ energy and degeneracy	CLM007	V005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2510.12880	Claim: open-chain $\theta=\pi/2$ degeneracy	CLM006	V004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Claim: periodic exact-point ground-state manifold	CLM001	V001	paper_scale_execution_not_run	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Claim: periodic phase and product-state controls	CLM002	V002				fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Claim: perturbation preserves the uniform-positive-w sector	CLM009	V007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Claim: $\theta=0$ fractionalized-state parity constraints	CLM008	V006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Main Figure 5(a)	CLM003	T001	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.12880	Supporting claim: ground-state overlap trends	CLM003	T001	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2510.26761	C01-THEOREM1-GME	CLM_C01.THEOREM1_GME	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C06-THEOREM2-TRACE-BOUND	CLM_C06.THEOREM2_TRACE_BOUND	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C07-THEOREM2-NEGATIVITY-IMPLIES-GME	CLM_C07.THEOREM2_NEGATIVITY_IMPLIES_GME	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2510.26761	C08-LEMMA2-BIPARTITE-POSITIVITY	CLM_C08.LEMMA2_BIPARTITE_POSITIVITY	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2510.26761	C09-LEMMA3-MULTIPARTITE-POSITIVITY	CLM_C09.LEMMA3_MULTIPARTITE_POSITIVITY	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C10-THEOREM2-CONVOLUTION-IDENTITY	CLM_C10.THEOREM2_CONVOLUTION_IDENTITY	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C11-KERNEL-REDUCED-WIGNER	CLM_C11.KERNEL_REDUCED_WIGNER	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C12-GAUSSIAN-KERNEL-CONSTRAINT	CLM_C12.GAUSSIAN_KERNEL_CONSTRAINT	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C13-COROLLARY1-NONCLASSICAL-DEPTH	CLM_C13.COROLLARY1_NONCLASSICAL_DEPTH	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C14-INTERFEROMETER-SUFFICIENT-GME	CLM_C14.INTERFEROMETER_SUFFICIENT_GME	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C16-COROLLARY2-FINITE-REGION	CLM_C16.COROLLARY2_FINITE_REGION	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C17-COROLLARY2-MONOTONICITY	CLM_C17.COROLLARY2_MONOTONICITY	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2510.26761	C20-COROLLARY3-PSD-WITNESS	CLM_C20.COROLLARY3_PSD_WITNESS	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C21-COROLLARY3-TRACE-BOUND	CLM_C21.COROLLARY3_TRACE_BOUND	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2510.26761	C22-COROLLARY3-MEASUREMENT-COST	CLM_C22_COROLLARY3_MEASUREMENT_COST	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2510.26761	C29-FIG1-STATE-NORMALIZATION	CLM_C29_FIG1_STATE_NORMALIZATION	T001				all_linked_items_finalized_by_fresh_review — none
2510.26761	C30-FIG1-NEGATIVITY-WITNESS	CLM_C30_FIG1_NEGATIVITY_WITNESS	T001				all_linked_items_finalized_by_fresh_review — none
2510.26761	C31-FIG1-SMOOTHED-ORIGIN	CLM_C31_FIG1_SMOOTHED_ORIGIN	T001				all_linked_items_finalized_by_fresh_review — none
2510.26761	F1-3D-CUT	CLM_F1_3D_CUT	T001				fresh_independent_review_inconclusive_after_attempt — none
2510.26761	F1-COM-WIGNER	CLM_F1_COM_WIGNER	T001				fresh_independent_review_inconclusive_after_attempt — none
2510.26761	F1-EQUAL-SLICE	CLM_F1_EQUAL_SLICE	T001				fresh_independent_review_inconclusive_after_attempt — none
2510.26761	F1-SMOOTHED-COM	CLM_F1_SMOOTHED_COM	T001				fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C01-PAULI-PROGRAMMABLE	CLM_C01_PAULI_PROGRAMMABLE	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C02-COVARIANT-PROGRAMMABLE	CLM_C02_COVARIANT_PROGRAMMABLE	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C03-COHERENT-NO-GO	CLM_C03_COHERENT_NO_GO	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C04-AD-NO-GO	CLM_C04_AD_NO_GO	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C05-H-HPTP-DIM-K	CLM_C05_H_HPTP_DIM_K	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C06-AD-HPTP-DIM-12	CLM_C06_AD_HPTP_DIM_12	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C07-QMATRIX-POLYTOPE	CLM_C07_QMATRIX_POLYTOPE	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C08-UNITARY-CONJUGATED-SEMIGROUP	CLM_C08_UNITARY_CONJUGATED_SEMIGROUP	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C09-COVARIANCE-GENERATOR-IFF	CLM_C09_COVARIANCE_GENERATOR_IFT	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C10-COVARIANT-QDS-CHOI	CLM_C10_COVARIANT_QDS_CHOI	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C11-COVARIANT-IRREP-UNITAL	CLM_C11_COVARIANT_IRREP_UNITAL	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C12-COVARIANT-FINITE-MIXTURE	CLM_C12_COVARIANT_FINITE_MIXTURE	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C13-GENERATOR-CHANNEL-CONDITION	CLM_C13_GENERATOR_CHANNEL_CONDITION	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C14-COMMUTATOR-CP-IFF	CLM_C14_COMMUTATOR_CP_IFT	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C15-COMMUTATOR-EXP-IFF	CLM_C15_COMMUTATOR_EXP_IFT	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C16-UNITARY-FAMILY-PROGRAMMABILITY	CLM_C16_UNITARY_FAMILY_PROGRAMMABILITY	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C19-INITIAL-COST-ZERO	CLM_C19_INITIAL_COST_ZERO	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C20-COST-MONOTONE-T	CLM_C20_COST_MONOTONE_T	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2512.08279	C21-ANALYTIC-CONTINUATION	CLM_C21_ANALYTIC_CONTINUATION	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C22-COST-FAITHFULNESS	CLM_C22_COST_FAITHFULNESS	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C23-NOISY-PROGRAM-BOUND	CLM_C23_NOISY_PROGRAM_BOUND	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C24-SCALING-INVARIANCE	CLM_C24_SCALING_INVARIANCE	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C25-COMMUTING-SUM-SUBADDITIVITY	CLM_C25_COMMUTING_SUM_SUBADDITIVITY	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C26-TENSOR-SUBADDITIVITY	CLM_C26_TENSOR_SUBADDITIVITY	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C27-TROTTER-APPROXIMATE-SUM	CLM_C27_TROTTER_APPROXIMATE_SUM	T006	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C28-CHOI-BLOCK-RETRIEVAL	CLM_C28_CHOI_BLOCK_RETRIEVAL	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C29-STEADY-STATE-POSITIVE-COST	CLM_C29_STEADY_STATE_POSITIVE_COST	T007	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C30-COHERENT-PROTOCOL-COST-2	CLM_C30_COHERENT_PROTOCOL_COST_2	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	C31-AD-CIRCUIT-KAPPA-5	CLM_C31_AD_CIRCUIT_KAPPA_5	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2512.08279	S-AD-HPTP-APPENDIX	CLM_C06_AD_HPTP_DIM_12	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2512.08279	S-AD-NOGO-APPENDIX	CLM_C04_AD_NO_GO	T004	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2512.08279	S-H-HPTP-APPENDIX	CLM_C05_H_HPTP_DIM_K	T005	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2512.08279	S-PAULI-APPENDIX	CLM_C01_PAULI_PROGRAMMABLE	T003	evidence_chain_incomplete	review_or_evidence_process_gap	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2512.23799	fig_stim_cirq_acceptance	CLM_T002	T002	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2512.23799	fig_stim_cirq_infidelity	CLM_T001	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2604.08669	F3-A-GNN-MAX	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2604.08669	F3-A-HUNGARIAN-MAX	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2604.08669	F3-B-GNN-AVERAGE	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2604.08669	F3-B-HUNGARIAN-AVERAGE	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2604.08669	Q-F3-AVERAGE-MEANS	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2604.08669	Q-F3-AVERAGE-NEAR-OPTIMAL	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2604.08669	Q-F3-MAX-GAP	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2604.08669	Q-F3-MAX-MEANS	CLM_FIG3_GNN_MOVEMENT_DISTANCE	T001	scientific_result_mismatch	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.15597	QCLM-S6-MQDT-CROSSCHECK	CLM_QCLM_S6_MQDT_CROSSCHECK	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	QCLM-S7-ANHARMONIC-TOGGLE	CLM_QCLM_S7_ANHARMONIC_TOGGLE	T027	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	QCLM-S8-R1-OPTIMAL	CLM_QCLM_S8_R1_OPTIMAL	T020	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	QCLM-S9-RF-SIGN-TOGGLE	CLM_QCLM_S9_RF_SIGN_TOGGLE	T028	scientific_result_mismatch	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-A-D1F12	CLM_S2_A_D1F12	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-A-D2F32-OPT	CLM_S2_A_D2F32_OPT	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-A-GUIDE-DCZ	CLM_S2_A_GUIDE_DCZ	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	attempted_failure_root_cause_system_capability_limit — none
2607.15597	S2-A-P0F12	CLM_S2_A_P0F12	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-A-P1F12-CURRENT	CLM_S2_A_P1F12_CURRENT	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-A-P2F32-OPT	CLM_S2_A_P2F32_OPT	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-A-S1F32	CLM_S2_A_S1F32	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-B-CURRENT-MISLABEL	CLM_S2_B_CURRENT_MISLABEL	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-B-D1P0-RATIO	CLM_S2_B_D1P0_RATIO	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S2-B-D2P2-RATIO	CLM_S2_B_D2P2_RATIO	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S5-A-MC-ATOM	CLM_S5_A_MC_ATOM	T024	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S5-A-MC-HYBRID	CLM_S5_A_MC_HYBRID	T024	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	S5-A-MC-ION	CLM_S5_A_MC_ION	T024	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	TS05-MQDT-POLARIZABILITY	CLM_TS05_MQDT_POLARIZABILITY	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	TS08-QLDPC-TRANSFER-MC	CLM_TS08_QLDPC_TRANSFER_MC	T020	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	TS09-PASSIVE-ION-MEMORY	CLM_TS09_PASSIVE_ION_MEMORY	T021	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	TS10-QLDPC-SE-MC	CLM_TS10_QLDPC_SE_MC	T022	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.15597	TS12-APM-BB-MC	CLM_TS12_APM_BB_MC	T023	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
2607.23978	CLM_COMPLETE_POVM_CFI_BOUND	CLM_COMPLETE_POVM_CFI_BOUND	T006	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2A.TH.A1.NH	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2A.TH.A2.H	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2B.TH.A1.NH	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2B.TH.A2.H	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2C.TH.AH	CLM_FRINGE	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2C.TH.ANH	CLM_FRINGE	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.23978	F2D.TH.AH	CLM_FRINGE	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2D.TH.ANH	CLM_FRINGE	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2E.TH.A1.IM	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2E.TH.A1.RE	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2E.TH.A2.DAG.IM	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2E.TH.A2.DAG.RE	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2E.TH.A2.IM	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2E.TH.A2.RE	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2F.TH.A1.IM	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2F.TH.A1.RE	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2F.TH.A2.DAG.IM	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2F.TH.A2.DAG.RE	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2F.TH.A2.IM	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F2F.TH.A2.RE	CLM_NONOPTIMAL_READOUT	T005	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F3A.TH.AH.VAR	CLM_ORDER	T003	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F3A.TH.ANH.VAR	CLM_ORDER	T003	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F3B.TH.AH.VAR	CLM_NOISE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F3B.TH.ANH.VAR	CLM_NOISE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F3C.TH.AH.DVAR.DGAMMA	CLM_NOISE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.23978	F3C.TH.ANH.DVAR.DGAMMA	CLM_NOISE	T004	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM01.LP.CSS.CONDITION	CLM.CLM01.LP.CSS.CONDITION	T018				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM02.LP.RATE.BOUND	CLM.CLM02.LP.RATE.BOUND	T018				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM03.CANONICAL.LOGICAL BASIS	CLM.CLM03.CANONICAL.LOGICAL BASIS	T025				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM04.MAGIC.PORT.DISTANCE	CLM.CLM04.MAGIC.PORT.DISTANCE	T019				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM05.MAGIC.INJECTION.DISTANCE	CLM.CLM05.MAGIC.INJECTION.DISTANCE	T019				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM06.CLASSICAL.QUANTUM.BOUND	CLM.CLM06.CLASSICAL.QUANTUM.BOUND	T020				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM07.FULL.RANK.COUNTEREXAMPLE	CLM.CLM07.FULL.RANK.COUNTEREXAMPLE	T028	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.28795	CLM08_COMMUTATOR_DISTANCE_BOUND	CLM_CLM08_COMMUTATOR_DISTANCE_BOUND	T020				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM09_ELEMENT_ORDER_BOUND	CLM_CLM09_ELEMENT_ORDER_BOUND	T020				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM10_2X4_DISTANCE_CEILING	CLM_CLM10_2X4_DISTANCE_CEILING	T020				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM11_SQETCH_TRIAL_VALIDITY	CLM_CLM11_SQETCH_TRIAL_VALIDITY	T026				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM12_SQETCH_SKETCH_INCLUSION	CLM_CLM12_SQETCH_SKETCH_INCLUSION	T026				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM13_SQETCH_GENERIC_PIVOT_HIT	CLM_CLM13_SQETCH_GENERIC_PIVOT_HIT	T021				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM14_RELATIVE_GROUP_PERMUTATION	CLM_CLM14_RELATIVE_GROUP_PERMUTATION	T022				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM15_TANNER_THICKNESS_LOWER_BOUND	CLM_CLM15_TANNER_THICKNESS_LOWER_BOUND	T023				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM16_MITTEN_THICKNESS_EXACT	CLM_CLM16_MITTEN_THICKNESS_EXACT	T023				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	CLM17_SPECTRAL_PLACEMENT	CLM_CLM17_SPECTRAL_PLACEMENT	T024				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_A_M150	CLM_FIG02_A_M150	T005	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_A_M200	CLM_FIG02_A_M200	T005	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_A_M300	CLM_FIG02_A_M300	T005	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_A_M540	CLM_FIG02_A_M540	T005	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_A_M975	CLM_FIG02_A_M975	T005	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_A_SURFACE	CLM_FIG02_A_SURFACE	T005	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_B_J300_X	CLM_FIG02_B_J300_X	T006	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_B_J300_XX	CLM_FIG02_B_J300_XX	T006	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_B_J300_XX10	CLM_FIG02_B_J300_XX10	T006	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_B_J500_ZZ	CLM_FIG02_B_J500_ZZ	T006	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG02_B_J540_XX	CLM_FIG02_B_J540_XX	T006	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG08_QDISTRND	CLM_FIG08_QDISTRND	T007	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG08_SQETCH	CLM_SQETCH	T003	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG09_TELESCOPING	CLM_FIG09_TELESCOPING	T008	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG12_2AOD_MITTEN	CLM_FIG12_2AOD_MITTEN	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG12_2AOD_STRUCTURED	CLM_FIG12_2AOD_STRUCTURED	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG12_4AOD_MITTEN	CLM_FIG12_4AOD_MITTEN	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
2607.28795	FIG12.4AOD_STRUCTURED	CLM_FIG12.4AOD_STRUCTURED	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG12.SLM.MITTEN	CLM_FIG12.SLM.MITTEN	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	FIG12.SLM_STRUCTURED	CLM_FIG12.SLM_STRUCTURED	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE01_DIMENSIONS	CLM_TABLE01_DIMENSIONS	T025				fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE01.DISTANCE_EST3	CLM_TABLE01.DISTANCE_EST3	T010	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE01.DISTANCE_EXACT5	CLM_TABLE01.DISTANCE_EXACT5	T009	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE01_GADGET_SUMMARY	CLM_TABLE01_GADGET_SUMMARY	T011	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2607.28795	TABLE01.LOGICAL_WEIGHTS	CLM_MITTEN	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE02.SX	CLM_TABLE02.SX	T011	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE02.SXX	CLM_TABLE02.SXX	T011	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE02.SY	CLM_TABLE02.SY	T011	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE02.SZ	CLM_TABLE02.SZ	T011	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE02.SZZ	CLM_TABLE02.SZZ	T011	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE03.PARALLEL_SURGERY	CLM_TABLE03.PARALLEL_SURGERY	T012	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE04_EXTRACTOR	CLM_TABLE04_EXTRACTOR	T013	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE06.ABLP_DISTANCE	CLM_TABLE06.ABLP_DISTANCE	T010	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE06.MITTEN_DIMENSIONS	CLM_MITTEN	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2607.28795	TABLE06.MITTEN_DISTANCES	CLM_TABLE06.MITTEN_DISTANCES	T009 — T010	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2607.28795	TABLE06.MITTEN_LOGICAL_WEIGHTS	CLM_MITTEN	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
2607.28795	TABLE06_STRUCTURED_DIMENSIONS_WEIGHTS	CLM_TABLE06_STRUCTURED_DIMENSIONS_WEIGHTS	T014	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE06_STRUCTURED_DISTANCE	CLM_TABLE06_STRUCTURED_DISTANCE	T010	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE08.CIRCUIT_DISTANCE	CLM_TABLE08.CIRCUIT_DISTANCE	T015	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE09_CONVERGENCE_THROUGHPUT	CLM_TABLE09_CONVERGENCE_THROUGHPUT	T008	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE11.MAX_SPEED	CLM_TABLE11.MAX_SPEED	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE11_TRANSFERS	CLM_TABLE11_TRANSFERS	T016	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
2607.28795	TABLE12.MITTEN_HAL	CLM_TABLE12.MITTEN_HAL	T017	indispensable_input_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none
PhysRevB.14.2239	Claim: irrational-field spectrum is a Cantor set	CLM_T008	T008	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
PhysRevB.14.2239	Claim: spectral continuity and measure discontinuity	CLM_T009	T009	implementation_not_ready	system_capability_limit	not_found_after_checks	at_least_one_required_target_attempt_failed — none
cond-mat-0503511	M2a.S1	CLM_LZF	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
cond-mat-0503511	M2a.S2	CLM_LZF	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
cond-mat-0503511	M2a.S3	CLM_LZF	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
cond-mat-0503511	M2a.S4	CLM_LZF	T002	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
physics-0206018	fig5_cross_section_curve	CLM001	T001	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
physics-0206018	fig6_near_field_intensity_map	CLM002	T002	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
physics-0206018	fig7_far_field_pattern	CLM002	T003	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
physics-9712001	table2_ground_state_asymptotic	CLM_NEAR_ONE	T004	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R015	CLM_COVARIANCE_COEFFICIENT_QUADRATURE	T004 — T006 — T007	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_supported_target — fresh_independent_review_inconclusive_target_after_attempt — none
quant-ph-0211074	R026	CLM_ISING_SCALING	T007	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
quant-ph-0211074	R027	CLM_ISING_SCALING	T007	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
quant-ph-0211074	R028	CLM_ISING_SCALING	T007	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
quant-ph-0211074	R032	CLM_XXZ_REGIME_DISCRIMINATION	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R033	CLM_XXZ_REGIME_DISCRIMINATION	T009	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R034	CLM_NONCRITICAL_SCALING_FUNCTION	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R035	CLM_NONCRITICAL_SCALING_FUNCTION	T010	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R037	CLM_NONCRITICAL_SATURATION	T001	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R046	CLM_CRITICAL_LOG_SCALING	T002	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R047	CLM_CRITICAL_LOG_SCALING	T002	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R048	CLM_INCREMENT_RATIO	T013	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — none
quant-ph-0211074	R049	CLM_FIG2_XXX_FINITE_CURVE	T002 — T014	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R052	CLM_FIG2_GUIDE_RESIDUALS	T002 — T011	scientific_result_mismatch	paper_claim_discrepancy	not_found_after_checks	fresh_independent_review_inconclusive_target_after_attempt — scientific_evidence_accepted — none
quant-ph-0211074	R053	CLM_DMRG_EFFECTIVE_RANK	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R054	CLM_DMRG_EFFECTIVE_RANK	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R055	CLM_DMRG_EFFECTIVE_RANK	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R056	CLM_DMRG_EFFECTIVE_RANK	T017	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0211074	R058	CLM_HIGHER_DIM_AREA_LAW	T018	parameter_values_unavailable	publication_underspecified	not_applicable	fresh_independent_review_inconclusive_after_attempt — none

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Table S5 continued

Paper	Check	Claim	Target	Direct cause	Root cause	Code fault	Reason / next action
quant-ph-0211074	R061	CLM_RG_MONOTONICITY	T015	parameter_values_unavailable	publication_underspecified	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0507004	exceptional_point_accumulation	CLM_T018	T018	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none
quant-ph-0507004	exceptional_point_exclusion_regions	CLM_T020	T020	implementation_not_ready	system_capability_limit	not_found_after_checks	fresh_independent_review_inconclusive_after_attempt — none

S1.5 Complete machine-readable ledgers

Supplementary Data provide one row per Claim Check in `checks.csv`, one row per Claim in `claims.csv`, and one row per paper in `papers.csv`. The complete objective-blocker and attempted-failure subsets are duplicated in `externally_blocked_checks.csv` and `attempted_not_reproduced_checks.csv`. Every row retains source location, Claim and Target identifiers, final disposition, direct cause, root cause, code-fault status, Fidelity state and next action. The machine-readable ledger is authoritative and does not truncate the 3,933-row denominator.